

Mouse germline cysts contain a fusome that mediates oocyte development

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eLife Assessment

This manuscript provides evidence that mouse germline cysts develop an asymmetric Golgi, ER, and microtubule-associated structure, referred to as Visham, which in many ways resembles the fusome of *Drosophila* germline cysts. This is an **important** study that provides new evidence that fusome-like structures exist in germ cell cysts across species. While most of the data are **solid**, several instances remain in which conclusions regarding the dynamics and function of Visham should be restated, or additional experimental evidence should be provided to more fully support the authors' interpretations.

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Abstract

Mouse female primordial germ cells undergo five synchronous, incomplete mitotic divisions, and send each resulting germline cyst into meiosis to fragment and produce 4-6 oocytes and 24-26 supportive nurse cells. How oocytes are specified, linked appropriately to nurse cells and acquire high quality organelles and cytoplasm remains unclear. We show that mouse cysts develop an asymmetric Golgi, ER and microtubule-associated structure, "Visham," similar to oocyte-determining fusome in *Drosophila* cysts. Starting as the EMA granule, Visham distributes asymmetrically in cyst cells along with Pard3, and enriches in future oocytes. Transient spindle remnants containing stable acetylated microtubules link early cyst cells; a gap in these microtubules beginning at the 8-cell stage suggests cysts are programmed to fragment into six-cell groups. Visham also acts during meiosis to mediate Pard3-dependent polarity, Xbp1-dependent adaptive UPR, and organelle rejuvenation. In *Dazl* mutants, cysts still form, but normal Visham behavior, polarity, rejuvenation and oocyte production are blocked.

Introduction

Early female germ cell development occurs within interconnected cysts formed by consecutive incomplete divisions of a germline stem cell (GSC) or PGC in diverse animals. These include basal species like hydra (Littlefield, 1994 [\[1\]](#)), nematodes like *C. elegans*, insects like *Drosophila*, and vertebrates including fish, frogs, mice and humans (Gondos, 1973 [\[2\]](#); King, 1970 [\[3\]](#); Pepling and Spradling, 1998 [\[4\]](#); Kloc et al. 2004a [\[5\]](#); Marlow and Mullins, 2008 [\[6\]](#); review, Spradling, 2024). As they develop, cysts acquire polarity often associated with a microtubule-rich "fusome" (Giardina, 1901 [\[7\]](#); Huynh and St Johnston, 2004 [\[8\]](#); Spradling et al 2022 [\[9\]](#)). In *Drosophila*, the fusome arises from mitotic spindle remnants, contains ER cisternae and its polarity is essential for oocyte production. Later in meiotic prophase the fusome mediates centrioles, mitochondria, and germ plasm regulators such as oskar mRNA to move into the oocyte where they form the Balbiani body (Bb). Remarkably, mammalian oocytes also undergo these same events (review: Spradling et al. 2022 [\[9\]](#)). Oocytes in many invertebrate and vertebrate species require Par genes to establish or maintain their polarity (review: Huynh and St Johnston, 2004 [\[8\]](#)).

In the mouse, germ cells are induced around embryonic day 6.5 (E6.5) (Lawson et al. 2002) and begin migrating toward the gonad as primordial germ cells (PGCs). En route, PGCs reprogram their epigenetic state and return to pluripotency (Seki et al. 2007 [\[10\]](#); Saitou et al. 2011 [\[11\]](#); Loda et al. 2022 [\[12\]](#); Liu et al. 2025 [\[13\]](#)). However, after they enter the gonad at E10.5, PGCs turn on DDX4 and initiate cyst formation while reprogramming is still ongoing. During E11.5 they upregulate Dazl, an RNA-binding, post-transcriptional regulatory protein which is needed to produce and translate multiple target mRNAs (Zagore et al. 2018 [\[14\]](#); Yang et al. 2020 [\[15\]](#)). Dazl plays an essential role promoting germ cell meiotic entry and oocyte production (Gill et al. 2011 [\[16\]](#); Nicholls et al. 2019 [\[17\]](#)).

Mouse germline cysts arise from five rounds of mitotic divisions beginning at E10.5, so that each PGC generates almost 32 cells. However, the cysts break into an average of 4.8 smaller cysts by the time divisions are completed at E14.5 when cells enter meiosis (Lei and Spradling, 2013 [\[18\]](#); Levy et al. 2024 [\[19\]](#)).

During meiosis, each smaller cyst slowly loses component cells, as every 2 days about one cell on average becomes activated to act as a nurse cell. It transfers most of its cytoplasm into the residual cyst, shrinks to a small remnant and undergoes programmed cell death (Lei et al. 2016 [\[20\]](#); Niu and Spradling, 2022 [\[21\]](#); Ikami et al 2023 [\[22\]](#)). In pachytene, there is a substantial reorganization that brings mitochondria and ER sheets into close proximity by E18.5 (Pepling and Spradling, 2001 [\[4\]](#)). Cyst breakdown is complete by 5 days after birth (P5) with the formation of 4-6 primordial follicles, each containing a single oocyte in which organelles transferred from nurse cells are gathered into an aggregate known as the Balbiani body. Cyst polarity guide cyst fragmentation and oocyte specification, polarization and organelle acquisition, but a mouse analog of the fusome has not been identified.

Germline cysts also support germ cell rejuvenation, the universal process that allows damage to be removed each generation so that species can persist without known limit. Germline rejuvenation takes place during meiosis in single celled eukaryotes (Unal et al. 2011 [\[23\]](#); Suda et al. 2024 [\[24\]](#); Xiao and Unal, 2025 [\[25\]](#)). Processes that rejuvenate organelles such as mitochondria, restore rDNA copy number, and reset epigenetic chromatin marks have also been documented in meiosis during animal gametogenesis (Cox and Spradling, 2003 [\[26\]](#); Hill et al. 2014 [\[27\]](#); Reik and Surani, 2015 [\[28\]](#); Bohnert and Kenyon, 2017 [\[29\]](#); Lieber et al. 2019 [\[30\]](#); Pang et al. 2023 [\[31\]](#); Yamashita, 2023 [\[32\]](#)). The Balbiani body, in newly formed oocytes, is associated with germ plasm formation and selection for mitochondrial function in many species (Heasman et al. 1984 [\[33\]](#); Kloc et al. 2004b [\[34\]](#); Marlow and Mullins, 2008 [\[6\]](#); Spradling et al. 2022 [\[9\]](#); Sekula et al. 2024 [\[35\]](#)).

Here we find that mouse PGCs and cyst cells elaborate a fusome-like structure we term "Visham" that is rich Golgi, endosomal vesicles and associates with ER. Visham distributes asymmetrically in association with stable microtubules that persist briefly for part of the cyst cell cycle. By the 8-cell stage, cyst cells reorganize into a rosette configuration that brings cells with the most bridges (potential oocytes) and associated Visham closer together. Developing cysts at this stage frequently fragment non-randomly at sites lacking microtubule connections into six-cell cysts, implying that mouse ovarian cyst structure is more highly controlled than previously realized. Beginning as PGCs and throughout cyst formation, EGAD-mediated UPR pathway proteins including Xbp1 are expressed, suggesting that the oocyte proteome quality begins to be enhanced beginning very early. After pachytene, centrosome/Golgi-rich elements move to the oocyte and mediate the acquisition of organelles by the Bb. In the absence of Dazl, cyst development and asymmetric Visham accumulation is disrupted, and no oocytes are produced.

Results

PGCs contain a Golgi-rich structure known as the EMA granule

Early observations of a germ cell "epithelial membrane antigen (EMA)" (Hahnel and Eddy, 1987) identified an "EMA granule" in mouse primordial germ cells that resembles the *Drosophila* spectrosome, a fusome precursor (Lin et al. 1994; Pepling and Spradling, 1998). To investigate this relationship further, we determined that the EMA granule appears in PGCs as early as E9.5 and continues to be expressed in later cysts (Figure 1A, Figure S1A). We used lineage labeling to mark cysts derived from single PGCs, and 3D surface rendering (Fig S1B-C) and volume quantification, to demonstrate a significantly uneven distribution of the EMA granule within cyst daughter cells (Fig 1B). This structural asymmetry prompted us to name the EMA-rich structure "Visham" (Sanskrit for asymmetric/uneven).

The EMA-rich material reorganized and moved closer together in female cysts by E13.5 (Figure 1C; Figure S1D-E), whereas in male cysts they remained mostly separate (Figure 1C' and graph). Cyst regions with enriched Visham staining contain an average of 3 Tex14-positive bridges, whereas unenriched zones had an average of just 1 (Figure 1D graph; Figure S1F). Thus, highly branched cells with more bridges, which are known to give rise to most oocytes (Lei and Spradling, 2013; Lei and Spradling, 2016; Ikami et al. 2023), contain more Visham. Similar clustering, known as rosette formation (Figure 1H), and increased fusome accumulation in pro-oocytes, occur as germline cysts grow in *Drosophila* and many species (see Hegner, 1914; Büning, 1994; Huynh and St Johnston, 2004). Rosette formation is thought to prepare for sharing between nurse cells and oocytes.

Visham is highly enriched in Golgi and vesicles

Electron microscopy (EM) of mouse germ cells documented EMA granules with prominent Golgi and endocytic vesicles in most E11.5 and E12.5 cyst germ cells (Figure 1E; Figure S1K). Visham co-stained with the Golgi protein Gm130 (Figure 1F) and the recycling endosomal protein Rab11a1 (Figure 1F').

Visham often partly overlaps with ER (Figure S1I). EM sections of E14.5 cysts revealed clusters of Golgi within adjacent cyst cells near ring canals as expected for cysts after rosette formation (Figure 1E, 1E'). EMA-1 antisera react with multiple fucosylated glycolipids (Apostolopoulos et al. 2015; Figure S1H). EMA staining disappears from germ cells at E14.5 (Figure 1I), however, very similar (but non-germ cell-specific) staining continued with wheat germ agglutinin (WGA) at later stages (Figure 1G, G'; Figure S1G).

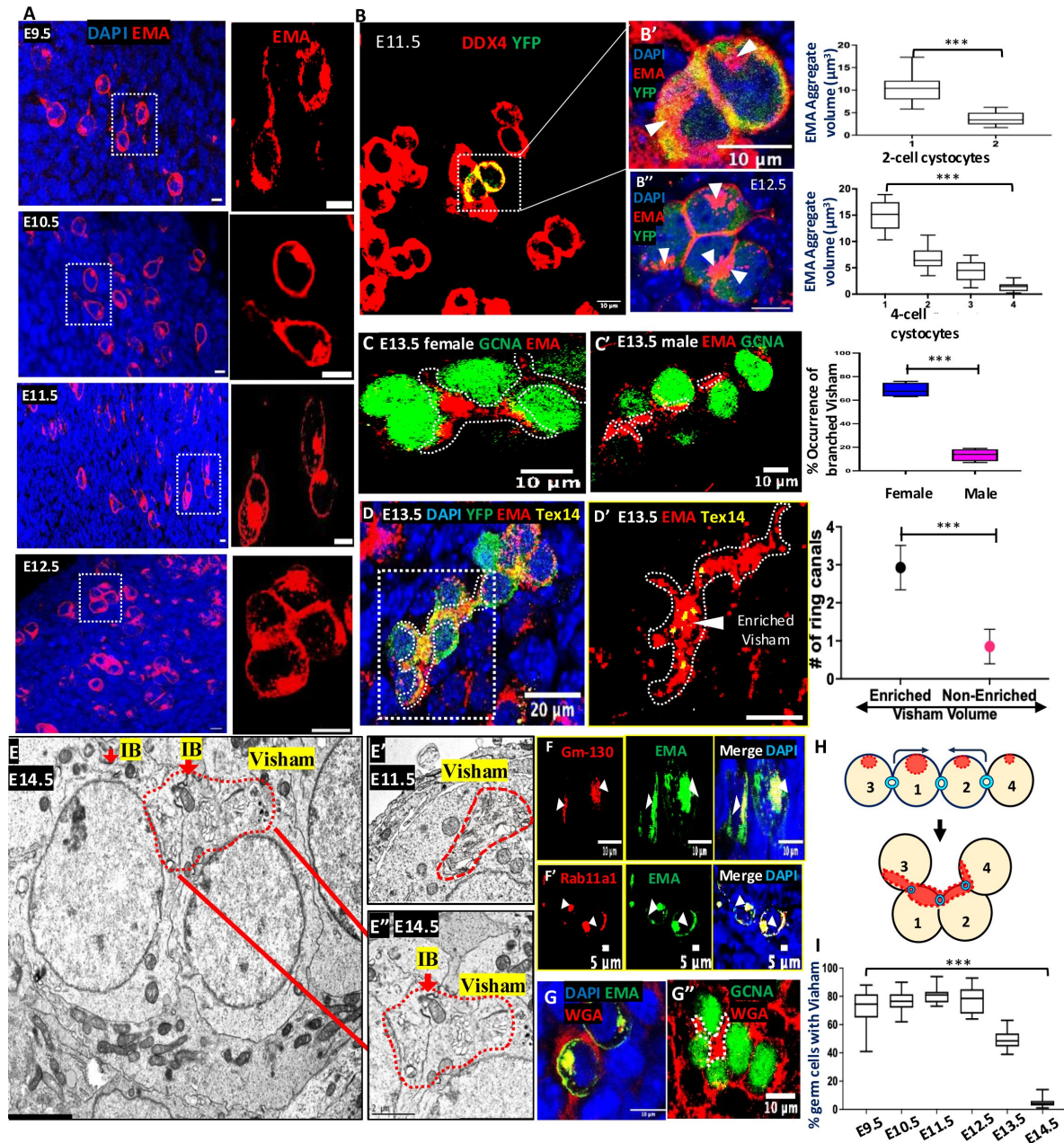


Figure 1.

Mouse pre-meiotic PGCs contain a fusome-like structure, "Visham."

A. E9.5-E12.5 ovaries: EMA, DAPI. Box: EMA granule (triangle) in germ cells. B. E11.5 ovary: lineage-labeled clones (YFP), DDX4. B'. 2-cell cyst. EMA granules (boxed). Graph: EMA volume asymmetry N=16. B''. 4-cell cyst: EMA granules (triangles). Graph: EMA volume asymmetry N=18; Student's t-test, ***p<0.001. C. E13.5 ovary and testis (C'): GCNA, EMA. Visham: (dotted line); Graph: % cysts with rosette Visham (N=26; ***p<0.001). D. E13.5 YFP-marked cyst: EMA, TEX14. D' (boxed region). Graph: Ring canal number vs. Visham enrichment ($\geq 10 \mu\text{m}^3$). (N=54; ***p<0.001). E-E''. EM images of rosette Visham spanning intercellular bridges: E14.5 (E, E''); EMA granule at E11.5 (E'). F. E11.5 ovary: GM130, EMA. F'. Rab11a1, EMA. G. E11.5 ovary: WGA, EMA; G'. E13.5 ovary: GCNA, WGA labels rosette Visham. (H) Schematic: Visham transformation from granular to rosette structure. (I) % germ cells with EMA structure during E9.5-E14.5 (N=15 per stage; ***p<0.001). Scale bars: 5 μm (A, F'), 10 μm (B-B'', C-C', D', F, G-G'), 20 μm (D), 2 μm (E).

Visham overlaps centrosomes and moves on microtubules

A cell's microtubule cytoskeleton that emanates from its centrosomes, and from non-centrosomal microtubule organizing centers (ncMTOCs), frequently controls trafficking of Golgi elements and positions them near centrosomes. To investigate the relationship between microtubules and Visham, we stained E11.5 gonads with anti-acetylated tubulin (AcTub) and EMA. Visham usually overlapped the centrosomes in interphase germ cells (**Figure 2A** [↗](#)), indicating it occupies a pericentriolar domain. In germ cells undergoing mitosis (**Figure 2B** [↗](#)), Visham associated with centrioles during interphase, but (typical for Golgi) dispersed as a spindle forms and for most of metaphase. In telophase, Visham vesicles gather at the spindle poles in both daughter cells and at the arrested cytokinesis furrow, which still retains the elongated spindle/midbody microtubules typical of cyst-forming cell cycles (**Figure 2B** [↗](#)). 3D summary movies of germ cells at these stages stained for AcTub and EMA are shown in **Figure S2A** [↗](#).

These behaviors differ in detail from the pattern of *Drosophila* fusome segregation during cyst divisions ([de Cuevas and Spradling, 1998](#) [↗](#)). Visham was rarely seen at just one pole of a spindle like the fusome at very early stages of GSC and cytoblast division; some EMA-reactive material also appears at the cytokinesis furrow earlier than the fusome (**Figure 2C** [↗](#); **Figure S2C-D** [↗](#)). Asymmetry in the Visham distribution arose only in late telophase. Clusters of Visham and centrosomes approached the arrested furrow (**Figure 2D** [↗](#), **S2C-D** [↗](#)). At this point short microtubule arrays were visible (**Figure 2D** [↗](#), **S2C** [↗](#)). Soon after, the amount of Visham in the two daughter cells becomes asymmetric, suggesting that differential activity or persistence of Visham transport late in cytokinesis generates asymmetry. Ovaries cultured briefly *in vitro* support continued germ cell division and Visham asymmetry. Visham formation is severely affected in such cultures if MTs were disrupted by cold or by Ciliobrevin D treatment (**Figure S2E graph** [↗](#)).

Cyst fragmentation is non-random and correlates with microtubule gaps

Mouse cysts produce an average of 4-6 oocytes per PGC as a result of initial cyst breakage into multiple, independently developing subcysts that each produce one oocyte ([Lei and Spradling, 2013](#) [↗](#); [Niu and Spradling, 2022](#) [↗](#)). Production of uniform-sized oocytes, as observed, would seem to require programmed breakage into uniform subcysts ([Spradling et al. 2022](#) [↗](#)), but some random breakage also occurs ([Lei and Spradling, 2013](#) [↗](#); [Ikami et al. 2023](#) [↗](#); [Levy et al. 2024](#) [↗](#)). To assess a role of microtubules and Visham in programming cyst fragmentation, we tracked AcTub and Visham in lineage-labeled E12.5 cysts. In about 10% of interphase cysts, presumably those in very late telophase or early in the subsequent cell cycle, persistent microtubule arrays were observed that connected the cells in pairs (**Figure 2E** [↗](#)). When we examined a total of about 50 large cysts, 15 larger cysts of 7-10 cells were found where the MTs in all cells could be analyzed completely. Unexpectedly, all showed a large discontinuity of 10 μ or more in microtubule linkage between just two of the cystocytes (**Figure 2E** [↗](#)). Three examples are shown in which projections of 8-cell lineage labeled cysts are outlined (dashed lines). In the left column, Visham and MTs are shown (**Fig. 2E-E'** [↗](#); **Figure S2F'** [↗](#)). In the middle column, microtubule bundles connect most cell cells, but are absent between two cells (straight dashed line). The absence of an MT connection predicts cyst breakage (**Figure 2E''** [↗](#)), where separation into a 2-cell and 6-cell cyst has already occurred.

The patterns of cyst breakage predicted by MT absence were non-random (**Figure 2F** [↗](#)). Production of a 6-cell cysts was strongly preferred based on the position of the gaps in 13/15 cysts analyzed. 8-cell or larger cysts all were predicted to generate a 6-cell cyst and a residual cell group. This pattern potentially explains six-cell mitoses frequently observed during E10.5-E14.5 ([Pepling and Spradling, 1998](#) [↗](#)), and the ~5-fold increase in the number of accumulated organelles in P4 oocytes compared to E14.5 germ cells ([Lei and Spradling, 2016](#) [↗](#)).

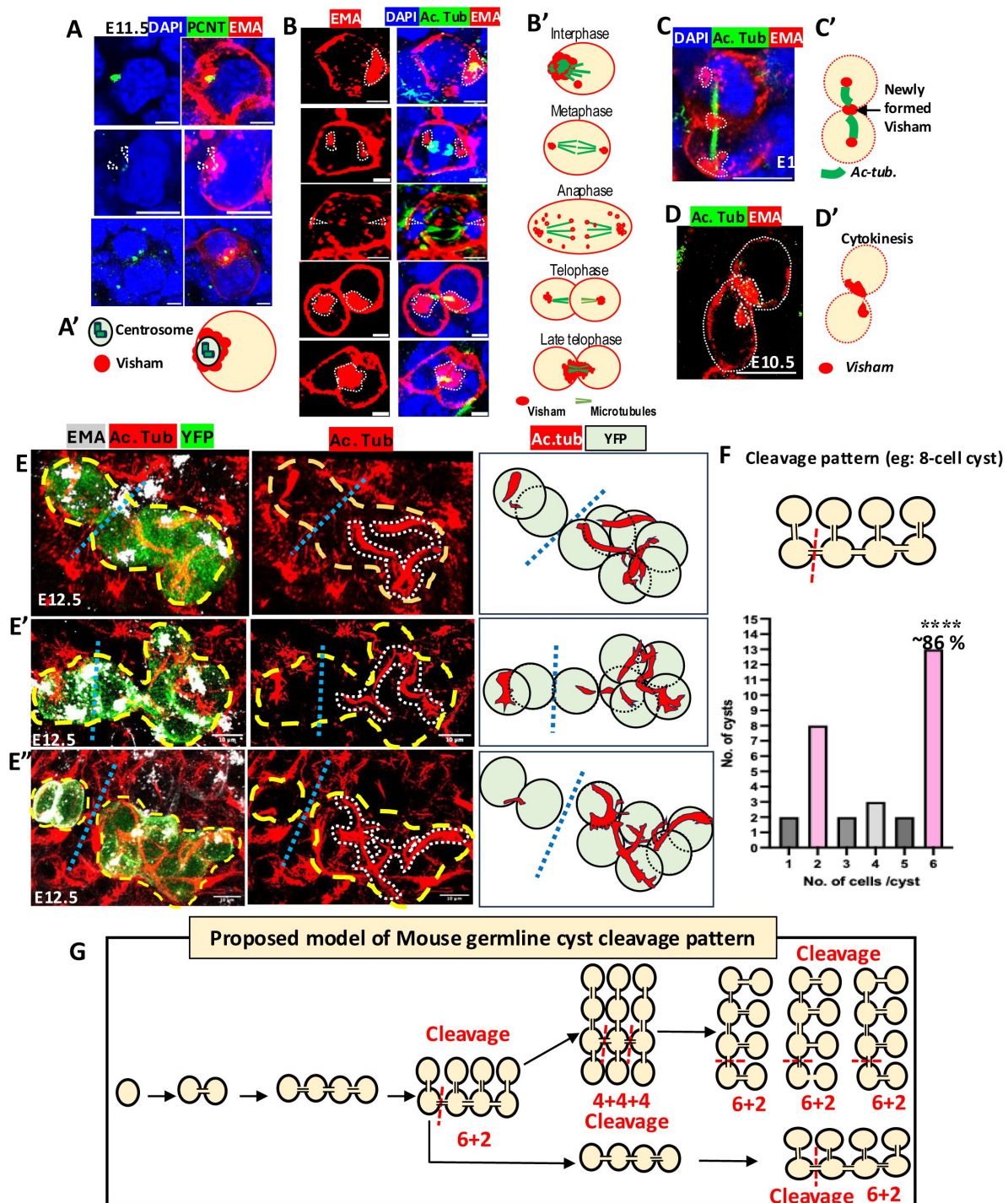


Figure 2.

Stabilized spindle microtubules mediate Visham asymmetry and cyst breakage.

A. Visham interphase pericentric localization. E11.5 germ cells: EMA, Pericentrin. A'. Summary. B. Cell cycle behavior of EMA during initial 2-cell cyst formation. B'. Summary diagrams. C-D. Asymmetric Visham segregation (E10.5). C. Symmetric telophase separation. D. Asymmetry at arrested cytokinesis. E-E''. Persistent spindle remnants presage cyst breakage. Three lineage-marked (YFP) E12.5 8-cell cysts (E-E'') in early interphase stained to reveal Visham and AcTub. MT channel alone shown in middle column. Diagrams at right suggest sites of breakage (dashed line). The E'' cyst has already broken into 2-cell and 6-cell cyst derivatives. F. Cell number distribution of cyst breakage products predicted from 15 lineage-labeled cysts analyzed as in E (7-cell: 3; 8-cell: 8; 9-cell: 1; 10-cell: 3). Binomial test: 6-cell dominance in 13/15 cysts (**** $p < 0.0001$). G. Model of PGC cyst production and breakage into four 6-cell cysts and 4 2-cell cysts. Scale bars: 5 μ m (B), 10 μ m (A, C-E).

Visham associates with polarity and microtubule genes including Pard3

Microtubules play a critical role in ovarian cyst formation and polarity in multiple species. Some of the most important regulators of MTs and polarity are the Par proteins that function in epithelial cell polarity, embryonic development, asymmetric neuroblast division (Petronczki and Knoblich, 2001 [\[1\]](#)) and oocyte formation in *C. elegans*, *Drosophila* and vertebrates (Kemphues et al. 1988 [\[2\]](#); Huynh and St Johnston, 2004 [\[3\]](#); Moore and Zernicka-Goetz, 2005 [\[4\]](#)). *Drosophila* meiotic cysts express Par3/Baz and Par6, another apical Par complex protein, along with beta-catenin/Arm in a ring-like arrangement close to the ring canals (Cox et al. 2001 [\[5\]](#); Huynh et al. 2001 [\[6\]](#); Huynh and St Johnston, 2004 [\[3\]](#)). This relatively small membrane zone likely represents the apical domain of cyst germ cells.

We investigated the expression of Pard3, the mouse ortholog of Par3/Baz to determine if a similar arrangement of the highly conserved Par proteins occurs in mouse germline cysts. Pard3 expression localized around Visham but extended beyond it, a pattern reminiscent of the larger ring of Baz in *Drosophila* cyst cells (Figure 3A [\[7\]](#), Figure S3A [\[8\]](#)). At E13.5, cysts in rosette configuration show Pard3 and Visham enriched around a central cell (Figure 3B [\[9\]](#), dashed region) but not in E13.5 male cysts (Figure S3B' [\[10\]](#)). This co-localization was validated in multiple lineage-labeled E13.5 cysts (Figure 3D [\[11\]](#); Figure S3D [\[12\]](#)). Ring canals were concentrated in this zone (Figure 3C [\[13\]](#)), hence mouse cysts have ring canals and an apical Par protein in the same small anterior location as in *Drosophila*, and that Visham occupies the corresponding zone as the *Drosophila* fusome. These findings suggest that mammals form and polarize oocytes using a conserved system found in diverse animals based on Par genes, cyst formation and a fusome.

Visham associates with Golgi genes involved in UPR beginning at the onset of cyst formation

We sequenced additional fetal germ cells (see Methods) and added them to data from Niu and Spradling (2022) [\[14\]](#) to make an integrated dataset of more than 3,000 germ cells spanning E10.5 to P5 (Figure 3F-H [\[15\]](#); Figure S3E-I [\[16\]](#)). Our gene expression studies showed that PGCs and early cysts actively express many genes that enhance proteome quality by monitoring, refolding or expelling misfolded or aggregated proteins for degradation. The UPR pathway associated largely with the ER in yeast is regulated by Hac1, the ortholog of Xbp1. We observed high levels of Xbp1 (Figure 4E [\[17\]](#)) and many of its target genes (Figure 4I [\[18\]](#)) that increase quality secretory protein production. Membrane proteins and lipids are also promoted by this pathway. In animals the UPR pathway has been expanded with several additional branches. We also observed expression Creb3l2 (ER/Golgi resident transcription factor), ATF4 (General UPR transcription factor activated by Golgi stress), Golp3 (Golgi phosphoprotein for membrane trafficking) and Arfgap(s) (involved in both Golgi to ER and Golgi to endosome trafficking) during mouse cyst formation (Fig S3J [\[19\]](#)). Xbp1 activates genes such as the chaperones (Hsp5a/Bip, Hsp90b1; DnaJc10; Lman2; DnaJb9, protein-disulfide isomerases (Erp44, Pdia3, Pdia4, Pdia6, Creld2), the ERAD regulator Syvn1/Hrd1 and the ERAD components (Sec61b; Ubxn4, Psmc6, Ube2j2, Get4, Ufd1). Thus, repair and expansion of ER, Golgi and secretory components begins immediately or even before migrating PGCs reach the gonad (Fig 3I [\[20\]](#)).

Xbp1 also supports synthesis of ER membrane phospholipids via upregulation of targets such as Elov11, Elov15, and Elov16, and Chka. Other highly expressed targets maintain ER and Golgi structure and are orthologs of *Drosophila* fusome proteins (Figure 3I' [\[21\]](#)). Thus the adaptive UPR pathway and its branches are active during cyst formation, and later in prophase as well (Figure 3I [\[22\]](#)-3I' [\[23\]](#)).

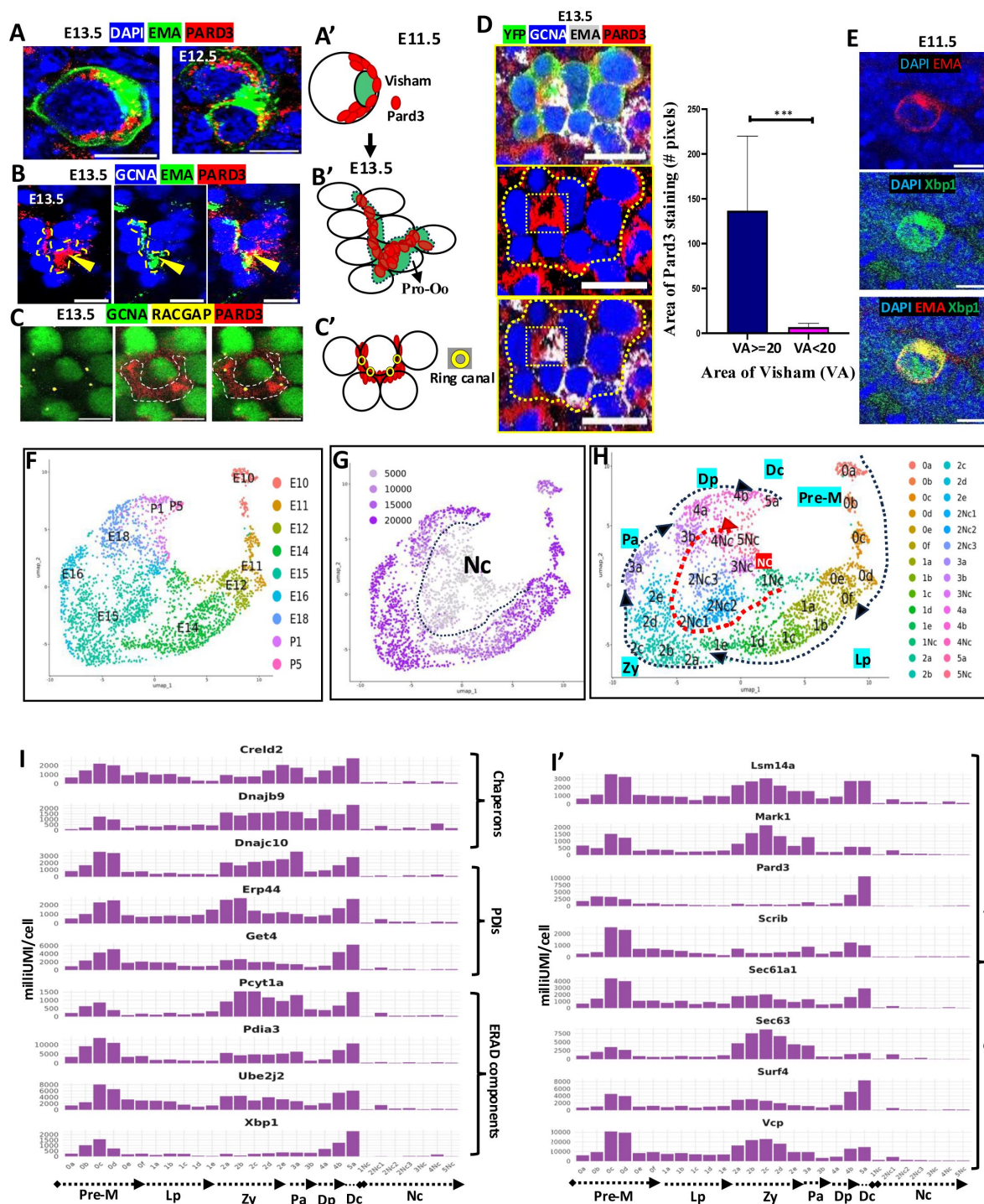


Figure 3.

Visham associates with Pard3 and apical polarity.

A-B. Pard3 overlaps Visham at E11.5-E12.5 (A, A'), and after rosette formation at E13.5 (B, B'). C-C'. Ring canals (RACGAP) localize within the Pard3+ apical domain. D. A lineage-labeled E13.5 cyst (YFP); channels below show overlap of Pard3 and Visham. (N=13; ***p<0.001). E. Xbp1 enrichment in EMA granule of E11.5 PGC. F-H. scRNAseq of E10.5-P5 germ cells. UMAP (F), UMI (G) Nc =nurse cells. Summary (H): pre-meiotic (Pre-M), leptotene (Lp), zygotene (Zy), pachytene (Pa), diplotene (Dp), dictyate (Dc). (I-I') Bar plots: Xbp1 and target expression plots. I'. Targets orthologous to fusome components. Scale bars: 10µm (A-C), 20µm (D).

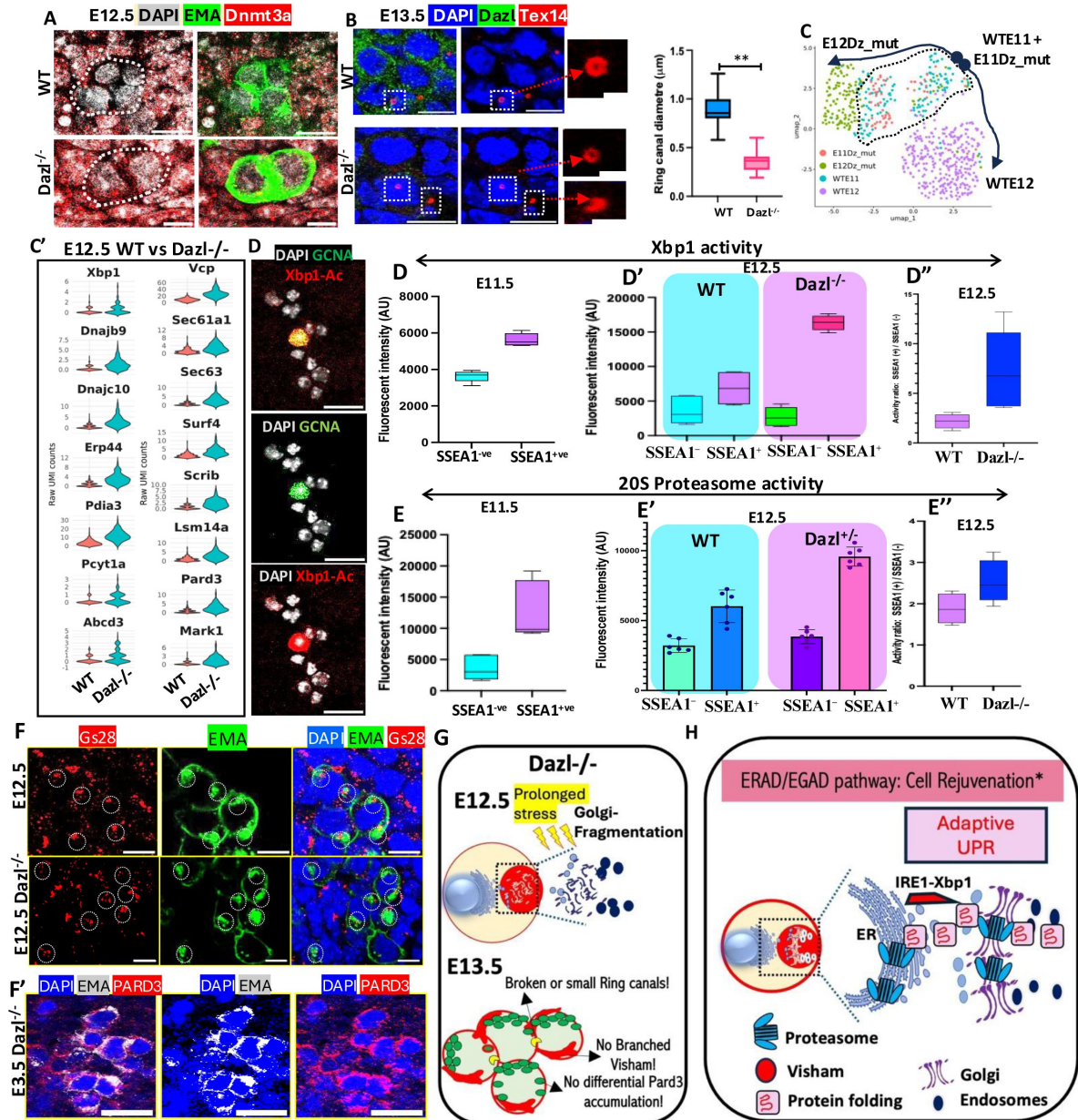


Figure 4.

UPR genes are active during cyst formation and controlled by Dazl.

(A) Dnmt3a and EMA levels at E12.5 are reduced in WT compared to Dazl^{-/-} germ cells. (B) Ring canals are smaller and defective in E13.5 Dazl^{-/-} cysts compared to WT. (N=44; **p<0.05). (C) Diverging expression of WT and Dazl^{-/-} germ cells in UMAP plots. (D-D'') IRE1-Xbp1 assay comparing SSEA1⁺ germ vs SSEA1⁻ somatic cells at E11.5 and WT vs Dazl^{-/-} germ cells at E12.5. (E-E'') Proteasome activity comparing SSEA1⁺ germ vs SSEA1⁻ somatic cells at E11.5 and WT vs Dazl^{-/-} germ cells at E12.5. (F) Golgi fragmentation in E12.5 Dazl^{-/-} germ cells. (F') Failure of E13.5 Dazl^{-/-} germ cells to form EMA rosettes or enrich PARD3. G. Dazl^{-/-} effects. H. ERAD-UPR proteostasis. Scale bar: 10µm (except zoomed in 2µm).

UPR genes are active during cyst formation and depend on Dazl

To investigate the effects of Dazl on cyst formation we confirmed that *Dazl*^{-/-} mice retard the PGC to germ cell transition (**Figure S4C-E**). *Dazl*^{-/-} E12.5 cysts showed higher levels of the DNA methylase Dnmt3a expression compared to controls (**Figure 4A**; **Figure S4A-B**) suggesting that germ cell DNA demethylation has slowed. Ring canals highlighted by staining for Tex14 were only half the size of wild type canals at E13.5 or structurally abnormal showing that cyst formation was slowed and cytokinesis was sometimes abnormal(**Figure 4B**).

We also performed scRNA sequencing of Dazl female homozygous mutant gonads at E11.5 and E12. and reclustered them with wild type germ cells at these stages for comparison (**Fig 4C**). At E11.5 the merged germ cells cluster together suggesting that *Dazl*^{-/-} has minimal effects on germ cells at E11.5, the stage it is first expressed (**Fig 4C**). However, E12.5 WT and *Dazl*^{-/-} germ cells cluster separately (**Fig 4C**). Wild type and *Dazl*^{-/-} expressed similar amounts of EGAD mediated UPR pathway genes at E11.5 germ cells (**Figure S4F**) whereas wild type germ cells expressed lower levels compared to *Dazl*^{-/-} cells at E12.5 (**Fig 4C'**). Thus, UPR genes in *Dazl*^{-/-} cells act similarly to pluripotency genes, and retain high expression levels later in cyst development than wild type.

The high expression of Xbp1 and many of its target genes in PGCs and early cyst cells argues that early germ cells are expanding ER and Golgi production, and increasing membrane protein quality. To validate that XBP1 is active in germ cells, we employed an IRE1-XBP1 ratiometric assay which utilizes a genetically encoded dual-fluorescent reporter system (see Methods, **Figure S4I**). First, we performed magnetic-activated cell sorting (MACS) to purify SSEA1 labelled germ cells from the E11.5 gonad (**Figure S4G-H**). We delivered the IRE1-XBP1 sensor to SSEA1+ve germ cells and SSEA1-ve control somatic cells by transfection. Both types of cells fluoresce in proportion to IRE1-splicing activity which measures Xbp1 activation activity. The construct also constitutively expresses a fluorescent marker such that the color ratio will represent a quantitative measure of relative IRE1-XBP1 activity between cells and under different conditions. Our observations showed that IRE1-Xbp1 activity in E11.5 germ cells was significantly higher than in somatic cells (**Figure 4D**). *Dazl* mutant cells showed persistent higher levels of Xbp1 activity at E12.5 (**Figure D', D''**).

The adaptive UPR pathway destroys misfolded proteins using proteasome activity, particularly the 20 proteasome (Vembar and Brodsky, 2008; **Figure 4H**). We measured proteasome activity within E11.5 PGCs to investigate whether the UPR pathway was paired with increased proteasome activity indicating it served an adaptive cellular mechanism of cell rejuvenation in PGCs destined to produce oocytes. Using a fluorogenic substrate to measure 20S Proteasome activity, we compared MACS sorted germ cells to somatic cells and found that germ cells have significantly higher proteasome activity than somatic cells at E11.5 (Fi 4E) and E12.5 (4E' Left). Proteasome activity was even higher in *Dazl*^{-/-} germ cells (**Figure 4E' Right, 4E''**).

We also studied the effects of Dazl mutation on the asymmetric enrichment of Visham in cells with multiple ring canals. Using both EMA and a Golgi marker, Gs28, we observed that germ cell Golgi begin to fragment in *Dazl*^{-/-} mutant germ cells even by E12.5 (**Fig 4F**, dotted circles). High levels of Golgi stress documented by the increased IRE1-Xbp1 activity, and elevated proteolysis in these cells, may lead to such fragmentation. These problems were accompanied by a dramatic arrest in cyst polarity development. E13.5 in *Dazl*^{-/-} germ cells showed virtually no differential expression of Pard3 or enrichment of Visham within particular cyst cells (**Figure 4F'**). Finally, in *Dazl*^{-/-} P0 gonads, large cells destined by become oocytes do not appear (**Figure S5E**). The effects of *Dazl*^{-/-} on germline cyst development and meiosis are summarized in **Figure 4G**. The observed loss of polarity caused by *Dazl*^{-/-} mutation can explain the failure of oocyte production and sterility of *Dazl* homozygotes.

Visham participates in organelle rejuvenation during meiosis

In *Drosophila*, starting at pachytene the microtubule cytoskeleton changes its structure, and polarity (Cox and Spradling, 2006). The new polarity, with microtubules running along the fusome such that minus ends cluster in the oocyte, prepares selected nurse cell organelles for transport into the oocyte and the Bb of a new primordial follicle. Mouse nurse cell centrosomes, mitochondria, Golgi, ER and possibly other organelles also move from nurse cells to the oocyte between P0 and P4 in a microtubule-dependent process (Lei and Spradling, 2016; Niu and Spradling, 2022). Prior to or during these events, *Drosophila* mitochondria are selected and rejuvenated by programmed mitophagy to enhance functionality (Cox and Spradling, 2003; Lieber et al. 2019; Palozzi et al. 2022; Monteiro et al. 2023). It is not currently known if selection and rejuvenation take place before movement to the oocyte, or if a significant amount also takes place en route.

Because the fusome is directly involved in organelle movement to the *Drosophila* Balbiani body we looked for changes in Visham and in gene expression that might relate to organelle rejuvenation and movement to the oocyte. Visham staining with WGA persists within meiotic germ cells at E17.5 (Figure 5A-A'; Figure S5A-C). Visham volume, remained in proportion with germ cell ring canals, as outer nurse cells (with one ring canal) transfer organelles and cytoplasm to cells with more ring canals located closer to oocytes (Figure 5A, graph). Pard3 content also showed such a proportion (Figure 5D). At E18.5, cyst development in the medulla toward wave 1 follicles, diverged from wave 2 follicles in the cortex that become quiescent primordial follicles (Figure 5B) (Yin and Spradling, 2025). Wave 1 oocytes grow substantially in volume, Visham content and Pard3 due to cytoplasmic transfer from surrounding nurse cells which decrease in size (Figure 5B-B', 5E-E', Figure S5E-F, left). Oocyte enrichment of Visham and Pard3 was confirmed using lineage labelled cysts at P0 (Fig 5C and 5F). Such enrichment was significantly reduced in *Dazl*^{+/-}-heterozygotes (Figure 5G', Figure S5E-F, middle).

Wave 2 follicles which produce nearly all fertile oocytes, increase organelle content beginning at E18.5 and contain a finished Bb by P4 (Niu and Spradling, 2022). Initially, 4-5 transferred Golgi-centrosome pairs were observed at separate cytoplasmic locations in oocytes. These gradually come together and form an oval-shaped cytocentrum that nucleates a microtubule aster and organizes the Balbiani body. We carried out immunofluorescent staining at E18.5 and examined EMs that documented interactions between Golgi, ER and mitochondria. Figure 5H shows a wild type germ cell with a large crescent of perinuclear mitochondria (AT5a) adjacent to a large cluster of Visham (WGA). An EM provides a higher resolution view of a similar interaction (Fig 5H'). In contrast, relatively few mitochondria or Visham were stained in a *Dazl*^{+/-}-heterozygote (Figure 5H'). Mitochondria also interact extensively with ER at this time (Figure 5I'; Pepling et al. 2007). ER tracker and Mitotracker (Figure 5I') stained clusters throughout much of the germ cell cytoplasm.

Discussion

Mouse germline cysts contain a fusome-like structure, Visham

Our results document a mammalian analog of the *Drosophila* fusome that displays conserved properties likely to underlie animal oogenesis. The fusome and Visham and both start as a small granular region within migrating PGCs before they arrive at the gonad. The granule transforms into an asymmetric structure during synchronous, incomplete divisions, involving stabilized spindle microtubules rich in AcTu and a program of arrested cytokinesis that produces distinctive ring canals. While stainable MTs persist for only part of the cell cycle, cyst and fusome polarity is maintained, as evidenced by preferential Visham accumulation in future oocytes, and rosette formation. Finally, the association of Visham with an apical Par complex located near the clustered

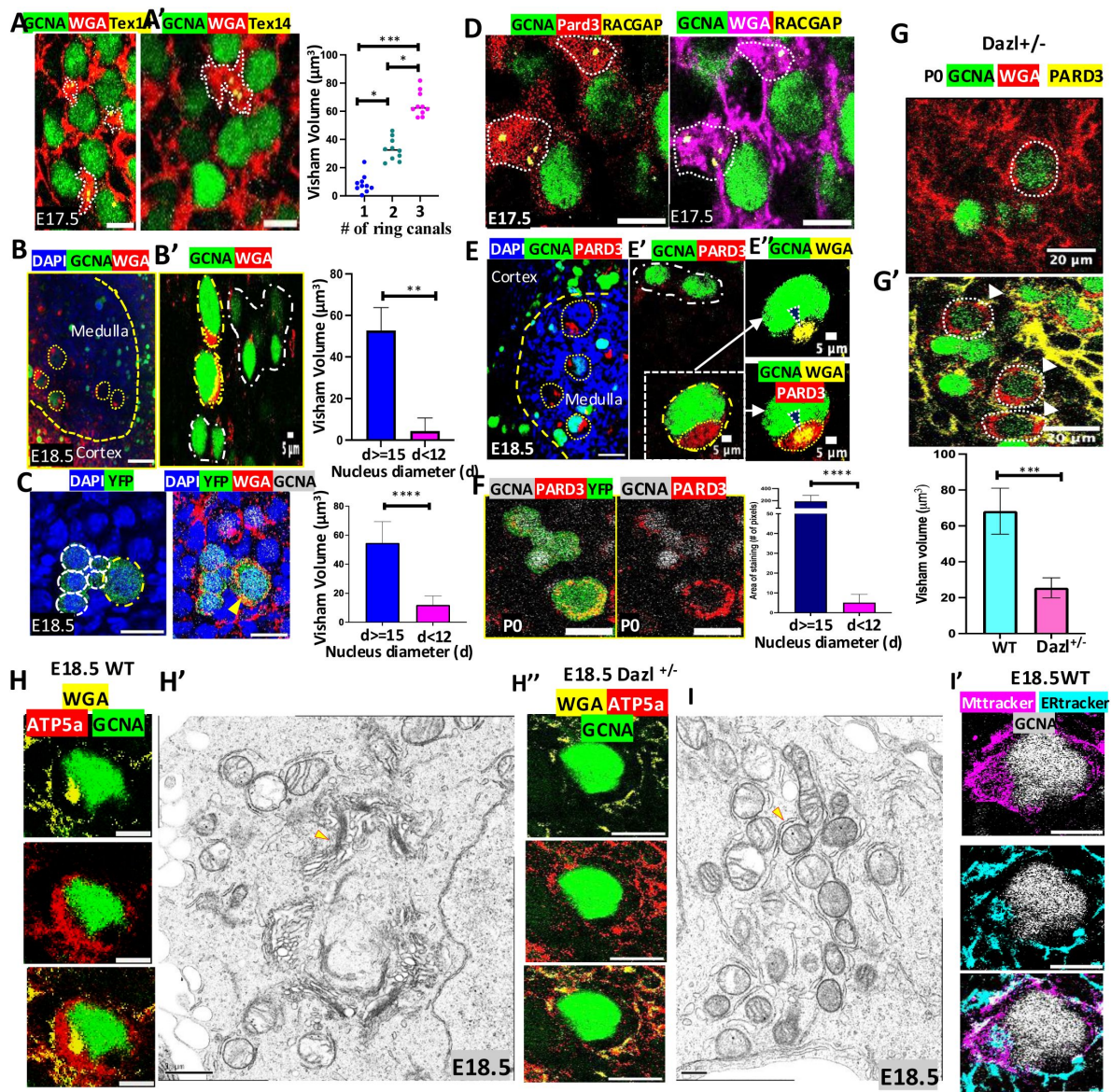


Figure 5.

Visham and Pard3 Associate with ER and mitochondria prior to Balbiani body formation.

A-A'. E17.5 ovary stained for WGA, GCNA, and TEX14 ; Graph: Visham volume versus ring canal number (N=65; ANOVA, * $p < 0.1$, *** $p < 0.001$). B-B'. E18.5 ovary shows Visham enrichment in large medullary oocytes vs smaller nurse cells; graph compares Visham volume versus Germ cell nucleus diameter (N=54; ** $p < 0.01$). Line: medulla/cortex boundary; dotted circle: large medullary oocytes; white dotted area: small nurse cells. C. Single-cell lineage-labeled E18.5 ovary shows Visham volume difference according to germ cell nucleus size (N=10; *** $p < 0.0001$). D. Pard3 enrichment along Visham proportional to number of ring canals. E-E'. Pard3 enrichment within newly arising medullary Oocytes (Big cells, GCNA) along with Visham (WGA). F. Lineage-labeled cells (YFP): Pard3 enrichment versus germ cell (GCNA) nucleus size. G-G'. Dazl^{+/-}-E18.5 ovary- Visham (WGA) and Pard3 enrichment failure in medullary Oocytes (GCNA). H-H". Organelle enrichment analysis: E18.5 (WT, H and Dazl^{+/-}-ovary, H") stained for WGA, mitochondrial marker ATP5a and GCNA. H' -EM image of Golgi-rich Visham (arrow) surrounded by mitochondria. I-I'. ER-mitochondria association in E18.5 ovary: I-EM image of ER tubules (arrow) wrapping mitochondria and I'-GCNA, ER and Mitochondria tracker staining. Scale bars: 20 μ m (A-F, G-G' H, I'), 5 μ m (B', E'), 0.5 μ m (EM images H', I').

ring canals, as originally described for the fusome in *Drosophila* cysts (Cox et al. 2001 [↗](#); Huynh et al. 2001 [↗](#)), is particularly important. Apical-basal patterning of *Drosophila* oocytes differs from many epithelial cells (review Huynh and St Johnston, 2004 [↗](#)), but Par genes are essential for oocyte production and their action now appears to be conserved at least in part. Our results argue that apical-basal polarity formation within germline cysts utilizes a fusome to specify oocytes and to aid their patterning and rejuvenation.

Early germ cells in other female mammals likely also use closely related mechanisms, given the ancient origin of cyst formation in animal gametogenesis. Golgi/ER-associated membranes forming a spherical mass in germ cells were reported in classical studies of pig and mouse reproductive tissues (Anderson & Beams, 1960 [↗](#); Clark & Eddy, 1975 [↗](#); Ginsburg et al., 1990 [↗](#)). Similar perinuclear structures associated with a Golgi complex in PGCs have been reported in other vertebrate species like chicken and quail (Fazel et al., 1987 [↗](#); 1990 [↗](#); Jung et al., 2005 [↗](#); Urven et al., 1988 [↗](#); Yoshinaga et al., 1991 [↗](#), 1992 [↗](#); Hen et al. 2014 [↗](#)). The presence of a fusome in developing human germline cysts and its similarity to mouse Visham merits further investigation.

Mouse cysts develop multiple uniform oocytes using a novel specific-cleavage mechanism

Mouse PGCs usually generate 4-6 oocytes using a complicated mechanism involving synchronous incomplete cyst divisions, limited breakage into smaller cysts (<10% of cell-cell junctions) and continuous, slow nurse cell turnover (Lei and Spradling, 2013 [↗](#); Lei and Spradling, 2016 [↗](#); Niu and Spradling, 2022 [↗](#)).

Despite their complex pathway, mouse oocytes in newly formed primordial follicles appear quite uniform in size and are efficiently produced, suggesting that they develop from final cysts of similar size and nurse cell content. However, lineage analyses did not reveal a pathway that generated uniform cysts, but showed many cyst cells that had moved considerable distances apart at least transiently. Live imaging studies revealed active movement likely responsible for some cell displacement, but also cell detachment caused by random breakage (Levy et al. 2024 [↗](#)).

Our studies of Visham and stable microtubules support the view that programmed cyst breakage during mitotic divisions contributes strongly to oocyte uniformity. AcTub arrays in cysts of 7-10 cells frequently contained a single large gap lacking MTs between a group of 6-cells, and the other residual cells. Such gaps in a major cytoskeletal element of germ cells subjected to significant tissue movements might represent cleavage sites (Figure 2E'' [↗](#)). Previous studies also strongly support the idea that 6-cell cysts play a special role. Synchronous mitoses in forming cysts are distinctive and easier to map unambiguously than the connectivity of lineage-labeled cells. 6-cell synchronous divisions are by far the most commonly observed cyst size not a power of 2 throughout E11.5 to E14.5 (Pepling and Spradling, 1998 [↗](#)). Additionally, quantitation of organelles in developing germ cells throughout oogenesis showed that P4 oocytes contain approximately 5 times as many centrosomes, Golgi, and mitochondria and four times as much cytoplasm as E14.5 germ cells (Lei and Spradling, 2016 [↗](#)). It is difficult to explain these results if random breakage due to excess random cell movement is a major factor determining cyst size and connectivity (Levy et al. 2024 [↗](#)), although it undoubtedly occurs.

What might cause the preferential breakage of growing cysts in a 6:2 manner? One possibility is the uneven distribution of Visham in four cell cysts (Figure 1B'' [↗](#)). The cell with the lowest amount might fail to sustain a microtubule connection at the next division, leading to 6:2 breakage. Involvement of the microtubule cytoskeleton in asymmetric divisions has been observed in many other situations (Fichelson & Huynh, 2007 [↗](#); Kaltschmidt & Brand, 2002 [↗](#); Sunchu & Cabernard, 2020 [↗](#); Meiring et al., 2020 [↗](#); Planelles-Herrero et al., 2022 [↗](#); Watson et al., 2023 [↗](#)). Even if programmed breakage into 6-cell cysts predominates, other sources of variation likely also occur. Some 8-cell cysts persist and divide again, rarely completing intact 32-cell cysts

(Pepling and Spradling, 1998 [↗](#)). It is not known if these cells are cleavable into smaller cysts or not, or whether random breakage to a smaller size would make them cleavable. Nonetheless, given oocyte uniformity, it seems plausible that most functional oocytes derive from cysts with about 5 nurse cells.

Germ cell rejuvenation is highly active during cyst formation

Germ cells have long been known to propagate species without know limits over evolutionary time by rejuvenating gametes each generation in association with meiosis (Weismann, 1892 [↗](#); Kirkwood, 1987).

Recently, the detailed cellular and molecular genetic mechanisms that make this possible both in single-celled eukaryotes and in animals have generated increasing interest (Cox and Spradling, 2003 [↗](#); Unal et al., 2011 [↗](#); Bohnert and Kenyon, 2017 [↗](#); Lieber et al., 2019 [↗](#); Chen et al. 2020; Palozzi et al., 2022 [↗](#); Spradling et al. 2022 [↗](#); Yamashita, 2023 [↗](#); Xiao and Unal, 2025 [↗](#)). By studying Visham, which is already present in migrating PGCs and E9.5, and by sequencing mouse germ cells from the earliest stages of ovarian development beginning at E10.5, we gained new insights into the timing and processes of germ cell rejuvenation during mouse oogenesis.

These studies made it clear that multiple processes of rejuvenation are highly active from the very earliest times of germ cell development (**Figure 3I** [↗](#)). In animals the UPR pathway has been modified by the addition of several additional pathways besides the Ire1-Xba1 branch, all of which were expressed in early germ cells. Creb3l2 controls a transcription factor that increases production COPII vesicles that carry out ER to Golgi transport and maintain a balance between secretory supply and demand. Insufficient matching can lead to Golgi stress (Machamer, 2015 [↗](#)), which can be managed by activating the ATF4 pathway branch.

Expression of Xbp1mRNA or even Xbp1 protein using a general antibody is not sufficient to show Xbp1 activity, since this is controlled by Ire1-mediated splicing to produce the active Xbp1s isoform. We were able to purify enough E11.5 and E12.5 germ cells to carry out biochemical assays and show that mouse female germ cells have substantial levels of Xbp1s activity at both times, higher than in somatic cells. The high expression of many known Xbp1 targets also documents the high activity of this pathway (**Figure 3I,I'** [↗](#)). Early germ cells also express genes suggesting that they are expanding membranes via lipid biosynthesis and secretory pathway activity. Consistent with active adaptive-UPR rejuvenation of their proteomes we documented substantial levels of active proteolysis in early germ cells at both E11.5 and E12.5

The observations reported here suggest that Golgi- and endosome-associated degradation (EGAD) — an emerging proteostatic mechanism plays a previously unappreciated role. EGAD complements the well-established ER-associated degradation (ERAD) pathway by facilitating the clearance of misfolded proteins via Golgi-endosome trafficking and proteolysis, rather than removal from the ER and degradation in the proteasome as in ERAD. We observed substantial amounts of ERAD based on the expression of Xbp1 and its many targets, and its documented activity and its endpoint 20S proteasome activity. It is hard to quantitatively compare the amount of EGAD activity, but the prominent role Golgi throughout the stages of mouse oogenesis and its location at the core of the Balbiani body suggests that EGAD either fulfills a unique role in oocyte production, or else is uniquely capable of operating at the highest level of proteostatic activity demanded by the much greater challenge of germline rejuvenation compared to any other tissue without generating damaging and self-limiting side effects.

Rejuvenation genes were also highly expressed later in meiosis during zygotene and pachytene, in diplotene when mitochondria/ER contact is extremely high (**Figure 5I** [↗](#)), as well as in primordial follicles (**Figure 3I,I'** [↗](#) "Dc"). The early appearance of Visham and its association with Golgi and ER-related processes that are central to adaptive-UPR, suggests that in animals, rejuvenation begins earlier in the germ cell cycle than it does in single celled eukaryotes. Animal germ cells return to a

pleuripotent state prior to undergoing cyst formation. Both these events precede meiosis and represent an addition to the rejuvenation process (Spradling, 2024). Our findings of early rejuvenation align with those of Palozzi et al. (2022) [\[1\]](#), who showed that mitochondrial rejuvenation during oogenesis uses a special germline mitophagy and begins at the onset of meiosis. In yeast, mouse and *C. elegans* oocytes, damaged materials are destroyed by lysosome-like degradation in mature oocytes and at meiosis II (Bohnert and Kenyon, 2017 [\[2\]](#); Zaffagnini et al. 2024 [\[3\]](#); Xiao and Unal, 2025 [\[4\]](#)). Thus, animal oocytes may engage in rejuvenation activities from before meiosis begins until it is completed.

Dazl regulates cyst formation, fusome polarization, and oocyte specification

Female gamete development in many organism is extensively controlled at the post-transcriptional level by RNA binding proteins (Mercer et al 2021 [\[5\]](#); Conti and Kunitomi, 2024 [\[6\]](#)). In vertebrates, the RNA-binding protein Dazl is widely conserved and controls multiple aspects of early germ cell development by regulating a large number of target transcripts (Zagore et al. 2018 [\[7\]](#)). In mice, Dazl modulates the ongoing reprogramming of new arrived gonadal PGCs to pluripotency beginning at E11.5 when it turns on (Gill et al. 2011 [\[8\]](#); Nicholls et al. 2019 [\[9\]](#)). Normally, Dazl continues to function at later stages of oocyte development (Conti and Kunitomi, 2024 [\[6\]](#)), but Dazl mutants fail to enter or progress in meiosis. In zebrafish, Dazl is required to establish germline stem cells and to properly form cysts via incomplete cytokinesis leading to their degeneration (Bertho et al. 2021 [\[10\]](#)).

Our studies defined a series of important new functions for Dazl during cyst formation. We confirmed that pluripotency reactivation is slowed in Dazl mutants since DNA demethylation was delayed, and pluripotency genes failed to start downregulating at E12.5. Dazl mutant homozygotes failed to normally regulate UPR genes, leaving their mRNA levels elevated. This likely resulted from a failure to downregulate their Xbp1 activator, since both Xbp1 activity and 20S proteasome activity was also elevated in Dazl^{-/-} germ cells. This overactivation of the UPR pathway appeared to cause Golgi fragmentation and These effects resembled the general slowing of normal developmental progression, as with pluripotency downregulation. We also observed abnormal cyst formation with some small and defective ring canals as i zebrafish Dazl mutants, but cysts remained intact and continued to develop. This allowed us to document that Dazl is needed for cysts and for the fusome to become polarized. In Dazl mutants, cysts did not undergo rosette formation or concentrate Visham or Pard3 in cells with multiple ring canals. Oocytes apparently could not be specified and did not form. These results show that Dazl acts as a major regulator that is essential for the most important aspects of cyst formation and oocyte production.

Methods

Resource availability

Lead contact Further information and requests for resources and reagents should be directed to and will be fulfilled by the Lead Contact, Allan C. Spradling (spradling@carnegiescience.edu).

Experimental model and subject details

Mouse strains used-C57BL/6, CAG-cre/ERT2 mice, R26R-EYFP (as described in Lei and Spradling, 2016 [\[11\]](#)) and Dazl1L^{+/-} mice (Strain #035880 from JACKSON laboratories).

Genotyping was performed according to protocols provided by the JAX Genotyping was performed according to protocols from the JAX Mice database. Animals were provided with a proper light-dark cycle, temperature, humidity, food and water. Sexually mature females and males (6-8 weeks

Reagent or Resource	Source	Identifier
Antibody/Lectin/Stains		
EMA	DSHB	EMA-1 AB_531885
GCNA	Abcam	ab82527
DDX4	Abcam/R&D systems	ab13840 & ab27591/ AF2030
SSEA1	DSHB	MC-480
WGA	Thermo Fisher	W7024
Gm-130	ThermoFisher/Novus Biologicals	610822/NBP2-53420
Rab9	Thermo Fisher	MA5-31997
Rab11a1	Thermo Fisher	700184
Tex14	Proteintech	18351-AP
GFP/YFP	Aves Labs	GFP-1020
Dazl	BioRad Labs/GeneTex	MCA2336/GTX89448
Lysotracker deep	Thermo Fisher	L7528
Pard3	Novus Biologicals	NBP1-88861
LAMP1	Cell Signaling Technology	9091T
Xbp1	Abcam	ab37152
Gs28	BD biosciences	61184
AAL	Vector Labs Inc	FL-1391-1
LCA	Vector Labs Inc	FL-1041-5
Sec63	Thermo Fisher	PA5-100180
Calnexin	Abcam	ab219644
Gs28	Proteintech	16106-1-AP
Acetyl- α -Tubulin (Lys40)	Cell Signaling. Tech	5335S
Alpha Tubulin (acetyl K40)	Abcam	ab289875
Pericentrin	Abcam	ab4448/ab28144
Anti-GCNA1 antibody	Abcam	ab82527
Rac GAP1 Antibody (A-6)	Santa Cruz Biotechnology, Inc	sc-271110
Dnmt3a	Cell Signaling. Tech	3598T
ATP5A	Abcam	ab14748
Mitotracker /Deep	Thermo Fischer	M7514/ M22426
ER-Tracker™ /	Thermo Fischer	E34251/ E34250
Quinacrine dihydrochloride	Sigma chemical	Q3251
Chemicals		

Key Resource table

Paraformaldehyde	Electron Microscopy Sci.	15714
Tamoxifen	Sigma Chemical	T5648
Corn Oil	Sigma Chemical	C8267
Tween 20	Sigma Chemical	P1379
Triton X	Sigma Chemical	X100
Mounting media	Vector Labs Inc.	H-1000
Trypsin-EDTA (0.25%)	Fisher	25200056
Fetal Bovine serum	Sigma Chemical	F2442
Bovine serum albumin	Sigma Chemical	A4503
SSEA1 (CD15) microbeads	Miltenyi Biotech	130-094-530
DMEM/F-12	Fischer	11320-033
Celiobrevin D	Millipore	250401
Critical Commercial kits		
Kappa fast hot start ready-mix	KAPPA Biosystems	KK5608
KAPPA express extract	KAPPA Biosystems	KR0383-v4.16
10X Genomics kit		
IRE1-Xbp1 Assay	Montana Molecular	U0921G
Proteasome Activity	Amplite	13456
Superscript IV cells direct cDNA synthesis kit	Fischer	11750510
Experimental models		
C57BL/6J	Jackson Laboratory	000664
CAG-Cre-ER	Jackson Laboratory	004682
R26R-EYFP	Jackson Laboratory	006148
Dazl1L ^{+/+}	Jackson Laboratory	035880
Others		
MACS MS separation columns	Miltenyi Biotech	130-042-201
Cell culture inserts	Millipore	PICM01250
Cell strainer	Sigma	CLS431752
Software and Algorithms		
FIJI	FIJI	Version: 2.16.0/1.54p
IMARIS	Oxford Instruments (formerly Bitplane)	Version: 10.2
SEURAT	Satija Lab	Version: 5.1.0
CELL RANGER PIPELINE	10X genomics	Version: 6.0.1/8.0.1
GRAPHPAD PRISM	GraphPad Software	Version: 10.5.0

Key Resource table (continued)

Primer Name	Sequence	Amplicon Size and Reference
Uba1_Foreward	5'-TGGTCTGGACCCAAACGCTGTCCACA-3'	Male- 198 bp Female- 217 bp <i>McFarlane et al., 2013</i> (https://doi.org/10.1159/0003486 ;
Uba1_Reverse	5'-GGCAGCAGCCATCACATAATCCAGATG-3'	
Sly_Foreward	5'-GATGATTTGAGTGGAATGTGAGGTA-3'	Male- 280 bp Female- 685 and 48 bp <i>McFarlane et al., 2013</i> (https://doi.org/10.1159/0003486 ;
Sly_Reverse	5'-CTTATGTTTATAGGCATGCACCATGTA-3'	
Zfy_Foreward	5'-GACTAGACATGTCTTAACATCTGTCC-3'	Male- 180 bp <i>McFarlane et al., 2013</i> (https://doi.org/10.1159/0003486 ;
Zfy_Reverse	5'-CCTATTGCATGGACTGCAGCTTATG-3'	
DDX4 Foreward	5'-GAGATTGCCTTCAGTACCTATGTG-3'	105 bp <i>Gao et al., 2011</i> (https://doi.org/10.1095/biolrepro.10.084590)
DDX4 Reverse	5'-GTGCTTGCCCTGGTAATTCT-3'	
GAPDH Foreward	5'-GGTGAAGCAGGCATCTGAGGG-3'	246 bp <i>Chunmei et al., 2011</i> (https://doi.org/10.1093/glycob/cv52)
GAPDH Reverse	5'-GGTGGGTGGTCCAGGGTT-3'	
Dazl Common	5'-GAC ATT ACT AAG AAA ACA GCA GTG G-3'	Mutant=~300 bp, Heterozygote=~300 bp and 192 b Wild type=192 bp <i>JAX Protocol (Strain #035880, Primer# 56340, 56341 and 56342)</i>
Dazl WT reverse	5'-TTC TGC ACA TCC ACG TCA TT-3'	
Dazl Mut Reverse	5'-ATC CCT CCC TTT AGG GCT CA-3'	

Table 1.

Primer details.

old) were kept for mating. Mating was confirmed through the observation of a vaginal plug. As per standard protocol, we designate the midday of the corresponding day as E0.5, which marks the beginning of embryonic development. Fetal gonads were dissected during E10.5-E18.5 as needed. The day pups are born is designated as P0 and were dissected for some experiments. The Institutional Animal care and use committee of Carnegie Institution of Washington approved procedures for animal handling and experimentation.

Method details

Single germ cell lineage labeling

The protocol used for single germ cell lineage labeling was performed as described previously (Lei and Spradling, 2016). Briefly, to obtain fetuses for lineage marking, adult female R26R-EYFP mice were mated with male CAG-cre/ERT2 mice. Tamoxifen was dissolved in corn oil (Sigma) and a single dose was injected intraperitoneally at 0.2 mg per 40 g body weight into pregnant female R26R-EYFP mice at E10.5. Fetal gonads/Pups were dissected as needed in between E11.5-E18.5/P0. To analyze the lineage-labeled germ cells, fixed and frozen whole gonads were subjected to whole mount staining for chicken-GFP, the number of lineage-labeled germ cells of each ovary was determined by examining optically sectioned Z-stack images generated by confocal imaging of the entire ovary.

Immunostaining

Dissected fetal gonads/ovaries were washed in Phosphate buffer saline (PBS) and fixed immediately in cold 4% paraformaldehyde overnight at 4°C. Following the fixation, the gonads were washed three times with PBST₂, which is a mixture of phosphate-buffered saline (PBS) with 0.1 % Tween 20 (Sigma) and 0.5 % Triton X (Sigma). Each wash cycle lasted for 30 minutes.

Next, gonads were incubated with primary antibodies mixed with Blocking solution (PBST₂+ 10 % Normal donkey serum) overnight at room temperature. Following day, gonads were incubated with fluorescein-conjugated secondary antibodies (Donkey Anti-Rabbit/mouse/rat Alexa-fluor 488/568/647, Invitrogen) overnight at room temperature. Next day, gonads were washed with PBST and stained with DAPI (Sigma) to visualize nuclei. Gonads were mounted on slides with a mounting medium (Vector Labs) and analyzed using confocal microscopy (STELLARIS 8 DIVE Multiphoton Microscope, Leica). Germ cell numbers are quantified in each ovary by manually counting EMA/DDX4 stained germ cells in multiple 0.45 µm optical sections from different ovaries.

3D reconstruction

Three-dimensional model images and movies were generated by Imaris software (Bitplane). For z-stack images, different visualization options in the Surpass mode of Imaris helped to gain visualization control of the objects. To perform surface rendering and volume quantification using Imaris software, we imported our 3D image dataset for EMA/WGA staining. We started with a "Volume" rendering for channels corresponding to EMA/WGA staining and then we adjusted the contrast, brightness and transparency in the "Display Adjustment" window. Next, we used the Surfaces module to create surface renderings by selecting the appropriate fluorescence channel, applied manual thresholding and adjusted smoothing parameters to define the structure accurately. Once the surface is rendered, we navigated to the statistics tab to access volume measurements. Further visualization of the results was performed in 3D or orthogonal views to confirm accuracy, and adjusted rendering properties such as color or transparency as needed. To create spots for easy counting of labelled cells, we used the "Spots" icon-option in the "Creation" menu and started the spot creation wizard. We set the approximate spot size as per the requirement. Next, we selected the "Manual Creation" option to place spots individually. Each click placed a single spot at the corresponding position.

Sexing and Genotyping

Tail samples from mouse were collected and subjected to lysis and DNA extraction using Proteinase K based lysis buffer treatment at 55°C overnight followed by 95°C incubation for 10 minutes. The samples were then centrifuged at 14000 rpm for 5 min to pellet the debris.

Extracted genomic DNA from supernatant was used directly and amplified with the Uba1, Sly and Zfy primer pairs for sex determination (McFarlane et al., 2013 [DOI](#), Table S2). PCR reactions were performed using KAPA Fast Hotstart ReadyMix with dye (Kappa Biosystems, #KK5608) using the following PCR parameters: initial denaturation at 95 ° C for 5 min, 35 cycles with 94°C for 30 s, 60° C and 72° C for 30 s, followed by final elongation at 72° C for 5 min. PCR products were analyzed with a DNA ladder (100 bp) on 2% agarose gels and visualized with ethidium bromide under UV-illumination. The male and female amplicon can be distinctly visualized due to different amplicon sizes of PCR products (See Primer details).

Electron microscopy

Whole ovaries were dissected in PBS and fixed in 4% Paraformaldehyde overnight. After three 3-min washes in cacodylate buffer, ovaries were postfixed in 1% OsO₄, 0.5% K₃Fe {CN}₆, in cacodylate buffer for 1 hour and were rinsed twice for 5 min in cacodylate buffer and once for 5 min with 0.05 M maleate (pH 6.0). The ovaries were stained in 0.5% uranyl acetate overnight at 4 degrees rinsed in water and dehydrated through an ethanol series. Following two 10-min washes with propylene oxide, the ovaries were infiltrated with resin. The resin-embedded specimen was polymerized by incubation at 45°C and 70°C for 12 hours each. Silver-gold sections were cut, stained with lead citrate, and observed in the electron microscope.

Single cell RNA sequencing of Wildtype E10.5, E11.5 and E15.5 gonad

Embryonic ovaries were dissected such that individual fetal gonad at E10.5 (18 gonads from 3 females), E11.5 (12 female gonad from 3 females) and E15.5 (10 gonad from 2 females) were collected. For E10.5 and E11.5: corresponding tail were labelled for identification and placed in 1× PBS on ice. We performed quick Quinacrine and DAPI staining (~15 min) of corresponding E10.5 and E11.5 fetal tail samples to eliminate the male gonad samples -with positive Quinacrine staining of tail samples. We then pooled the dissected female gonad sample and dissociated it into single cells using 0.25% Trypsin at 37°C for 5-7 min with two pipet trituration in between. Fetal bovine serum (10 %) was then used to neutralize trypsin.

Dissociated cells were passed through 100-µm strainer. The cell suspension was centrifuged at 300 g for 5 minutes and cell pellet was resuspended in freshly prepared and filter-sterilized 0.04% BSA. Viability of cells was assessed via trypan staining and >90 percent viable samples were selected and loaded (~10000 live cells for E11.5 and E15.5, ~20000 cells for E10.5) onto the 10X Genomics Chromium Single Cell system using the for E11.5 and E15.5-v3 and for E10.5 v4 chemistry as per the manufacturer's instruction. Single-cell RNA capture and library preparations were performed, and standard data was data processing was performed using Cell ranger pipeline (6.0.1 - for E11.5 and E15.5 and 8.0.1-for E10.5) and later data was visualized and analyzed by Seurat v5.1.0.

Single cell RNA sequencing of Dazl1L mutant gonad

Dazl^{+/−}-female and males were kept for mating and plugged females were marked as E0.5. The gonad from E11.5, E12.5 fetuses were dissected, fetuses were collected in and kept in cold PBS in 12-well plate such that each well with individual fetus is marked for identification. Fetal tails were collected, labelled and subjected to kapa express extract kit (Catalog #KR0383-v4.16) mediated fast DNA extraction according to manufacturer's protocol. Briefly, one step lysis and DNA extraction system was set up in 100 µl volume by adding 10 µl express extract buffer, 2 µl of express extract

enzyme and 88 µl of PCR grade water in each tail sample followed by 75⁰ C incubation in heating block for 15 minutes for lysis. The samples were then incubated at 95⁰C in heating block for 5 minutes for DNA extraction. The sample was then centrifuged briefly to pellet the debris, and supernatant was used directly for PCR as mentioned in methods section of RNA isolation, cDNA synthesis and PCR. The standard genotyping JAX protocol for strain 035880 is refer to identify homozygous / heterozygous and wild type fetuses.

Simultaneously male/female sexing was also performed (primer details in Table S2). After correct identification, homozygous fetuses-from E11.5 (6 gonad from 3 females) and E12.5 (6 gonad from two females)-were trypsinized to prepare live single cells which were then subjected to 10X Genomics Chromium (v3 chemistry) and data was processed using cell ranger pipeline 6.0.1 and analyzed using seurat v5.1.0 in same manner as mentioned previously for E11.5 gonad.

Cell Identification and Clustering Analysis

Single-cell RNA sequencing (scRNA-seq) data were analyzed using the Seurat package (v5.1.0, <https://satijalab.org>). Count data generated by the Cell Ranger pipeline (6.0.1/8.0.1) were imported into R using the Read10X function and converted into a Seurat object with the CreateSeuratObject function. R packages were used to filter out the low-quality cells, and the following criteria were used to filter cells:

1. **for E10.5 WT gonad:** nFeature_RNA > 500 & nFeature_RNA<8500 & nCount_RNA > 500 & percent.mt<5
2. **for E11.5 WT gonad:** nFeature_RNA > 100 & nFeature_RNA<11000 & nCount_RNA > 300 & percent.mt<10
3. **for E15.5 WT ovary:** nFeature_RNA > 500 & nFeature_RNA<9000 & nCount_RNA > 500 & percent.mt<10.

The filtered count matrices were Log-normalized using the “NormalizeData” function and scaled with the “ScaleData” function to prepare for dimensionality reduction. “Principal Component Analysis (PCA)” was then applied to reduce dimensionality, followed by using top 15 dimensions and default resolution to cluster cells based on gene expression profiles using the “FindNeighbors” and “FindClusters” functions. Cell populations were visualized using the umap method, facilitating the identification of distinct cell types.

Bioinformatic segregation of female germ cells

To segregate E10.5 and E11.5 XX-specific germ cells with complete surety we performed bioinformatic segregation of female germ cells to address the limitations of primitive Quinacrine staining. The expression of the X-linked gene *Xist*, *Ddx3x*, *Utx* etc., were used as a marker. Cells with significant *Xist* or other X-linked gene expression were labeled as “XXonly” using the “Identify” and “WhichCells” R functions. A new Seurat object containing only XX-specific cells was created using the subset function. Germ cells within this subset were identified by examining the expression of germ cell-specific marker genes, and a further subset of E10.5 and E11.5 female germ cells was created using the subset function to focus on germ cells specifically.

Wildtype ger cells-merged dataset creation and Validation

ScRNA-seq datasets from E11.5 to P5 (retrieved from the GEO database, GSE136441 and merged object submitted to Github) were visualized using Seurat. E11.5 cells were discarded and new merged dataset from E10.5-P5 was created, subsets of germ cell clusters from E10.5, E11.5, and E15.5 were integrated using Seurat’s “Merge” function. The resulting merged dataset was then processed by normalizing and scaling the data, followed by the identification of variable genes. The batch correction was performed using and pre-harmony and post-harmony data was assessed by visualizing LISI (Local Inverse Simpson’s Index) score confirming successful batch mixing. Cell

clusters were determined using Seurat's single nearest neighbor (SNN) algorithm with PCA action using top 15 dimensions and 2.8 resolution. To visualize these clusters, dimensionality reduction was applied (umap) with following criteria: RunUMAP(new_merge, dims=1:15, n.neighbors=40, min.dist=0.6, spread=1) enabling the identification of distinct cell populations.

Cell cycle stages of germ cells across the different developmental stages (E10.5-P5) were validated by visualizing meiosis stage specific expression pattern via feature plot. This visualization technique allows for the capture of subtle trends and helps in comparing expression peaks, providing insights into the transition from mitosis to meiosis across the different developmental stages. Additionally, a violin plot was employed to visualize the variability in average raw UMI levels across germ cells. This approach allowed for an effective comparison of gene expression profiles, highlighting the differences in gene expression patterns between different stages.

Dazl mutant merged dataset

For E11.5 and E12.5 *Dazl*^{-/-} dataset, cells were filtered based on following criteria:

1. **for E11.5 *Dazl*^{-/-} ovary** : nFeature_RNA > 500 & nFeature_RNA<9000 & nCount_RNA > 500 & percent.mt<5
2. **for E12.5 *Dazl*^{-/-} ovary** nFeature_RNA > 500 & nFeature_RNA<10000 & nCount_RNA > 500 & percent.mt<5

Top 15 dimensions were used at default resolution to cluster the cells which were then visualized by umap. To create merged dataset from subset containing only germ cells were created and merged using “merge” function as described previously. Three dataset were created -for E11.5 wildtype and *Dazl*^{-/-} dataset, for E12.5 and *Dazl*^{-/-} dataset and one dataset was created where both E11.5 and 12.5 WT and *Dazl*^{-/-} were merged. Batch correction was applied using harmony. Merged dataset was normalized, scaled and top 15 dimensions at resolution of 0.4 was used for visualization using umap.

In vitro gonad culture: microtubule inhibitor treatment and organelle tracker assay

To perform *in vitro* culture of mouse fetal ovaries on membrane inserts with or without inhibitors, we prepared sterile culture medium, constituting DMEM/F-12 supplemented with 10% fetal bovine serum and penicillin-streptomycin. Inhibitor stock solutions were prepared in DMSO and diluted at desired concentration in culture media (Ciliobrevin D from Millipore, 25 μ M for 6 hours). Control medium containing the vehicle DMSO was prepared simultaneously. Fetal ovaries were dissected from E11.5 embryos under a stereomicroscope in sterile PBS, we have kept the surrounding mesonephric tissue intact and attached to the gonad to ensure proper development. We have also collected corresponding fetus tail of each fetal gonad to perform genetic sex determination. We have placed membrane inserts into a 24-well plate and added ~500 μ L of culture medium (with or without inhibitors) to the lower chamber and ensure no contact between medium and the insert surface. Next, we gently placed isolated ovaries onto the membrane surface using sterile forceps. The plate is then incubated in a CO incubator at 37°C for ~6 hours. After the designated culture period, confirmed ovaries (post genetic-sex analysis), were carefully removed from the membrane inserts and washed three times with basal media to remove any residual inhibitors. Next, we collected ovaries and fixed in 4% PFA for downstream immunolocalization studies. To inhibit the microtubule growth using cold treatment, we transfer dissected gonad to a chilled culture medium, maintained at 0-4°C. The tissue is fully immersed in the pre-cooled medium and incubated on ice for ~60 minutes. This low-temperature exposure destabilizes microtubules by disrupting tubulin dynamics, effectively depolymerizing existing microtubules and inhibiting further polymerization. After the incubation, the tissue is washed with fresh ice-cold medium and fixed in 4% PFA. To stain specifically for organelles: Fetal ovaries were dissected and cultured *in vitro* in DMEM/F12 medium supplemented with organelle-specific fluorescent dyes. LysoTracker™ Deep

(Thermo Fisher, L7528, 1 mM stock) or MitoTracker™ (Thermo Fisher, M7514) or MitoTracker™ Deep (Thermo Fisher, M22426), and ER-Tracker™ (Thermo Fisher, E34251) or ER-Tracker™ (Thermo Fisher, E34250) were each added at a final concentration according to the manufacturer's recommendations. The ovaries were incubated in this dye-containing medium at 37 °C with 5% CO for 12 hours. Following incubation, tissues were collected and fixed in 4% paraformaldehyde (PFA) for subsequent immunostaining.

Magnetic activated cell sorting (MACS)

Embryonic ovaries were pooled and treated with Trypsin then were neutralized followed by centrifugation as mentioned before to in ScRNA Sequencing method section to create dissociated single cells. The resulting cell pellet was re-suspended in ~80 microliters MACS buffer i.e. PBS with 0.5% BSA and 2 mM ethylene diaminetetraacetic acid (EDTA). ~20 microliters Anti-SSEA1 (CD15) microbeads (MiltenyiBiotec Inc., #130-094-530) were added to the cell suspension followed by incubated for 20 min on ice. After adding 1 ml MACS Buffer to the suspension, the SSEA1 bound cells were pelleted by centrifugation at 300 g for 10 min at 4°C. SSEA1+ve and SSEA1-ve cells were then separated by applying cell suspension to MS columns according to the manufacturer's instruction (MiltenyiBiotec Inc.). The SSEA1+ve cells were retained by the column and were filtered thrice to obtain pure germ cell population. The cells were counted, and viability was assessed via trypan staining and using automated cell counter (Fischer Scientific). For validation of MACS protocol the cells were fixed in 4 % PFA to perform Immunostaining of DDX4 and were also stored in trizol to perform RNA synthesis and PCR for germ cell specific gene expression.

IRE1-Xbp1 Assay

The IRE1-XBP1 ratiometric assay (from Montana Molecular #U0921G) is a genetically encoded biosensor system utilizes a BacMam Vector carrying dual-fluorescence biosensor. For each assay reaction 50 µl transduction mix consisting of XBP1-IRE1 sensor (15 µl) plus sodium butyrate (0.6 µl per well) was prepared in basal media. To standardize the working conditions, manufacturer provided thapsigargin (1 µM per well) was used as positive control and untransduced cells were used as negative control. For each experiment equal amounts of SSEA1-ve and SSEA1 +ve cells were added to a 96-well plate in duplicates. ~50 microliters of transduction mix was added to each reaction well. The plate was incubated for 45 minutes at room temperature and then transfer to 5% CO₂ and 37°C for 24 hrs. The enzymatic activity was then measured at 37°C by monitoring XBP1-IRE1 sensor's fluorescent intensity (top read) at Ex/Em=488/525 nm and basal constitutive fluorescent intensity at 565/620 using microplate reader (BioTek Synergy H1). IRE1-Xbp1 assay comparing MACS sorted SSEA1 germ vs SSEA1 somatic cells at E11.5 was performed across 6 experiments (~32 mice, ≥5 mice and ≥20 ovaries per experiment). IRE1-Xbp1 assay comparing SSEA1 vs SSEA1 E12.5 cells in WT or Dazl mutant mouse across 3 experiments (~40 mice, ≥5 mice and ≥25 ovaries per experiments)

20S Proteasome activity assay

To perform proteasome activity assay (Amplite #13456) we followed manufacturer's instructions. We first standardized the working conditions using trypsin enzyme as technical positive control. Trypsin is loaded on different well in different concentrations in duplicates. Blank medium along with assay buffer were added in separate wells to act as Negative control. For each experiment equal amounts of SSEA1-ve and SSEA1 +ve cells were added to a 96-well plate in duplicates. ~50 microliters of working solution consisting of fluorogenic substrate in assay buffer (prepared according to manufacturer's instructions) were added and plate was incubated for 2 hours at 37°C. The enzymatic activity was then measured at 37°C by monitoring fluorescent intensity (top read) at Ex/Em=490/525 nm (Cut off =515 nm) using microplate reader (BioTek Synergy H1). Proteasome activity in MACS-sorted SSEA1 vs SSEA1 cells were assessed using 3 biological assays with ~35–60 E11.5 ovary per assay. Proteasome activity ratio of MACS-sorted SSEA1 and SSEA1 cells compared at E12.5 with ~25–28 E12.5 ovaries were used per assay.

RNA isolation and cDNA synthesis

Total RNA was extracted using TRIzol reagent (Invitrogen) following the manufacturer's instructions. Briefly, tissue samples were homogenized in TRIzol, or cells were incubated in TRIzol for 5 min at RT, followed by phase separation with chloroform (200 μ l/ml TRIzol, shake for 10-15 seconds and keep at RT for 5 min) and centrifugation at $12,000 \times g$ for 15 minutes at 4°C. The aqueous phase was collected, and RNA was precipitated with equal amount of isopropanol (10 min RT then same centrifugation as last step), washed with 75% ethanol, centrifuged, removed the supernatant ethanol and air-dried. Resuspended pellet in RNase-free water. RNA quality and concentration were determined spectrophotometrically. For cDNA synthesis we used Superscript IV cells Direct cDNA synthesis kit (Invitrogen, #11750150) with and modified manufacturer's protocol. We skipped the lysis step and directly added DNase I to our sample (0.5 μ l each) followed by 10 minutes incubation on ice. Then we added Stop solution (3 μ l each) and kept the mixture at RT for 2 minutes. Then we set up RT reaction as per manufacturer's instruction by adding 8 μ l of RT mix to each sample and setting up the following reaction in PCR thermocycler: 25°C for 10 min, 55°C for 10 min, 85°C for 5 min and hold at 4°C. cDNA was used directly for PCR as mentioned in Sexing and genotyping using DDX4 and GAPDH Primers (See Primer details).

Quantification and statistical analysis

Quantification of germ cell numbers

PFA fixed ovaries were subjected to whole mount immunostaining as described previously. Z-stack images for whole ovaries were analyzed using FIJI. Germ cell numbers are quantified in each ovary by manually counting EMA/DDX4 stained germ cells in multiple 0.45 μ m optical sections from different ovaries using FIJI.

3D volume Quantification

Whole mount immunostaining was performed and the z-stack images were converted to 3D image to analyze using Imaris as described previously. After using the Surfaces module to create surface rendering and applying manual thresholding and smoothing to define the structure accurately, we navigated to the statistics tab to access volume measurements. Further visualization of the results was performed in 3D or orthogonal views to confirm accuracy, and adjusted rendering properties such as color or transparency as needed. On an average the average values from three measurements performed were considered and plotted as graph using Graphpad prism.

Area of staining quantified using FIJI

Images were analyzed using FIJI to calculate area of staining by selecting region of interest and using measure command from Analyze tab. The results were plotted as number of pixels to depict the area corresponding to stained regions.

Statistics

Data are presented as mean $\pm$ SEM (Standard error of mean). Data were analyzed by t-test/one-way Anova. P-value less than 0.05 is denoted by one star. P-value less than 0.01 is denoted by two stars. P-value less than 0.001 is denoted with three stars.

Data availability

Raw sequencing data has been deposited at Gene Expression Omnibus (GEO) database (accession: GSE303512). Any additional information required to reanalyze the data reported in this paper is available upon request.

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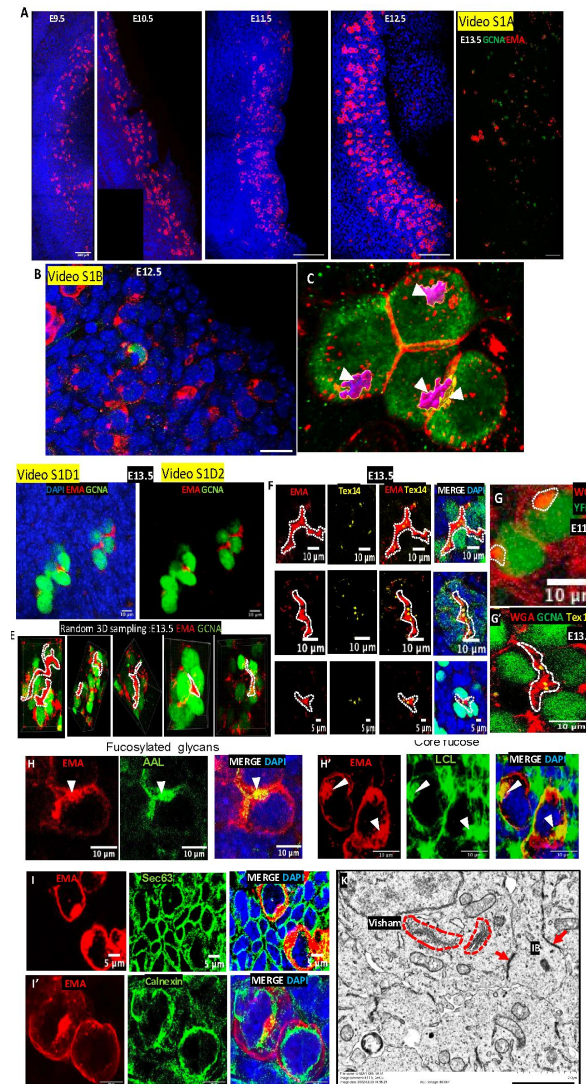
Supplemental information

Supplementary Videos: S1A, S1B, S1D1, S1D2, S2A1, S2A2, S2A3, S2A4, S2A5, S2F, S3A, S3B1, S3B2, S5C, S5D, S5E1-E3, S5F1-F3 can be found here: https://drive.google.com/drive/folders/1XuOoesa_WKMJUaVTpOKGbWWWhn_fjc56M

Fig. S1.

EMA/Lectin-stained aggregate (Visham) distribution in pre-meiotic PGCs

(A) Immunostaining of E9.5-E12.5 ovary for germ cell specific marker EMA (Red) and DAPI (Blue) and for E13.5 ovary (EMA in Red, GCNA in green, Z-stack video see Video S1A). (B) Z-stack video of E12.5 ovary stained for EMA (Red), DAPI (Blue) and lineage labelling shown via YFP (Green) (See Video S1B) (C) Volume rendering of EMA aggregate (White arrows) within lineage labelled cyst (Green, YFP) using Imaris software. (D-D') Videos of at E13.5 ovary stained for EMA (Red), GCNA (Green) and DAPI (Blue) (see Video S1D1 and S1D2) (E) Random 3-D sampling using Imaris for E13.5 ovary showing branched Visham structure within GCNA labelled female germ cells. (F) Images of different E13.5 ovary stained for EMA (Red), GCNA (Green), Tex14 (Yellow) and DAPI (Blue). Central region with enriched Visham is often associated with higher number of ring canals. (G) E11.5 ovary stained for WGA (Red) and YFP (Green) as lineage labelling marker. Dotted line marks the WGA aggregate within germ cells. (G') E13.5 ovary stained for WGA (Red), GCNA (Green) and Tex14 (yellow) forming branched WGA-stained Visham structure (dotted lines). (H and H') E11.5 gonad stained for general fucosylation specific lectins, AAL/LCA (Green), DAPI (Blue) and Visham marker EMA (Red). (I and I') Localization of ER markers in vicinity to Visham is validated by staining ER-specific Sec63 or Calnexin (Green), DAPI (Blue) and EMA (Red). (K) Electron microscopy of E11.5 gonad depicting Golgi clusters (Dotted red lines, labelled as Visham) near intercellular bridges (IB marked by red solid arrows). Scale bar. 100 μ m (A), 20 μ m (B), 10 μ m (in D-H).



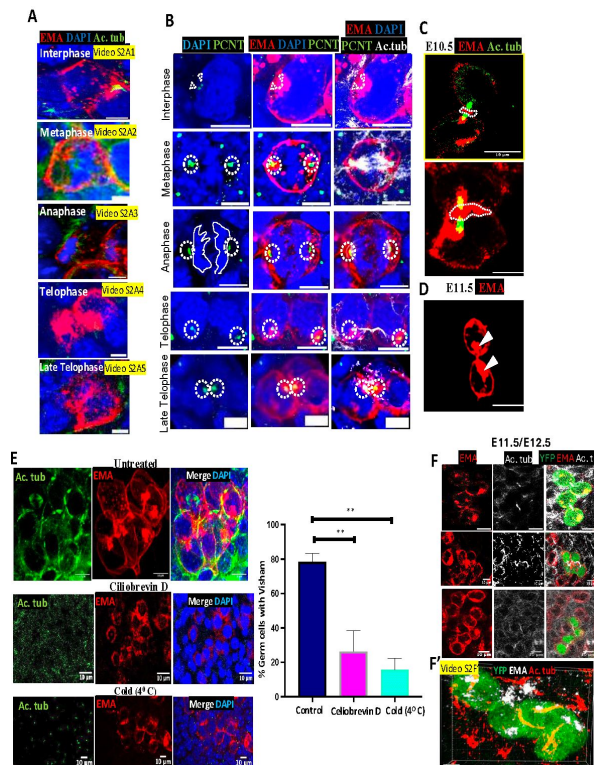


Fig. S2.

Validating microtubule dependent Visham formation and its distribution during cyst fragmentation.

(A) Z-stack videos for images used in [figure 3A](#) (See Videos from S2A1 to S2A5). (B) E11.5-E12.5 ovary stained for EMA (Red), Pericentrin (PCNT, Green), Ac. Tub (Grey) and DAPI (Blue). Arrows mark duplicated centrosome during Interphase. Dotted circles mark centrosomes. Smooth line marks the DAPI stained region during Anaphase depicting clear separation of Nuclei. (C) E10.5 gonad stained for EMA (Red) and Ac. Tub (Green). Dotted line marks the newly formed fusome as spindle remnant. (D) Visham asymmetry arises during cytokinesis depicted in E11.5 ovary stained for EMA (Red). (E) In-vitro cultured untreated gonad and treated gonad (microtubule inhibitor Ciliobrevin D and cold treatment) stained for Ac. Tub (Green), EMA (Red) and DAPI (Blue). Quantification showing % germ cells with Visham in Untreated versus Celiobrevin D/Cold treated E11.5 gonad. (F) Lineage labelled 4-cell cyst stained at E12.5 for YFP (Green), EMA (Red) and Ac. Tub (Grey). (F') 3D modelling of lineage labelled cyst at E12.5. Video depicts animation video generated using Imaris software to create uniform spheres in green to position the labelled 8-cells within cyst and surface rendering of Visham in White within lineage labelled germ cells (See Video S2F). Scale bar: 5 µm (A-B), 10 µm (C-F).

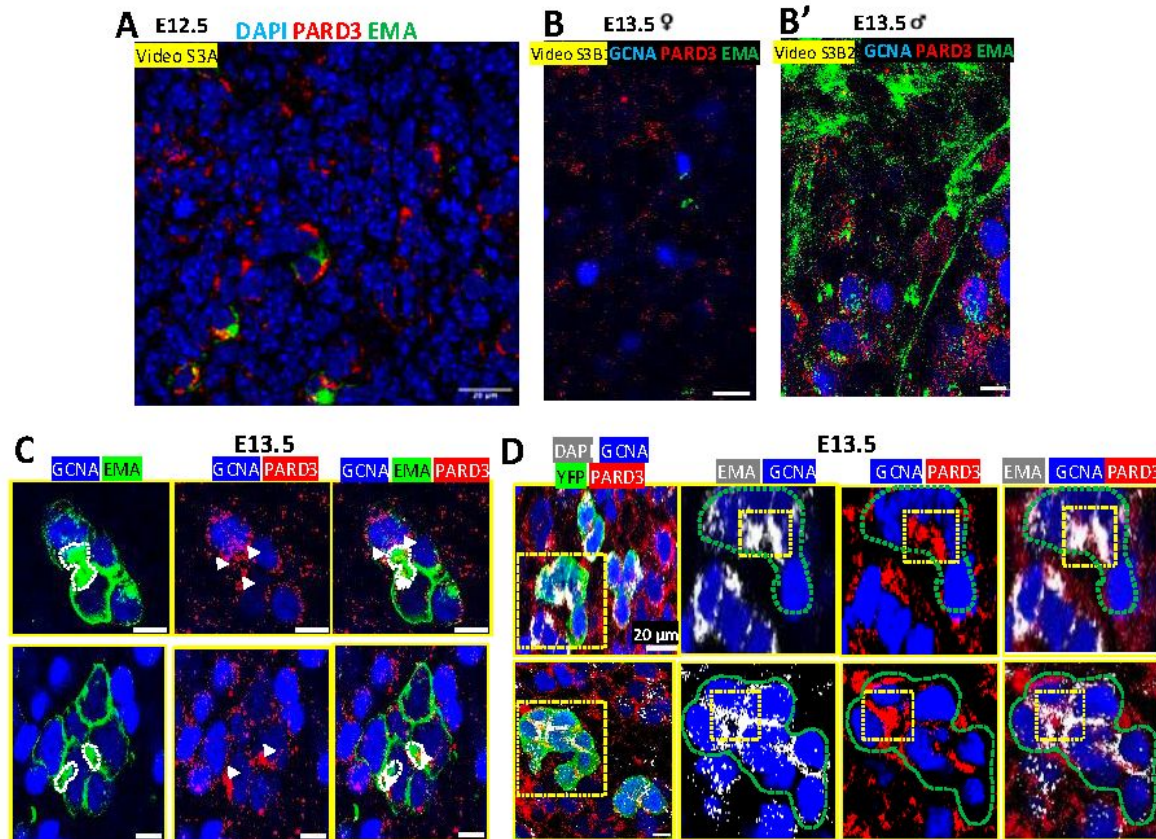


Fig. S3.

Pard3 gene expression in E12.5-E13.5 gonad

(A) Z-stack video of E12.5 gonad stained for EMA (Green), PARD3 (Red) and DAPI (Blue) (See Video S3A). (B) Z-stack video of E13.5 ovary (♀, See Video S3B1) and E13.5 testis (♂, See Video S3B2) stained for EMA (Green), PARD3 (Red) and GCNA (Blue). (C) E13.5 ovaries stained for GCNA (Blue), EMA (Green) and PARD3 (Red). (D) Lineage labelled E13.5 ovary stained for YFP (Green), GCNA (Blue), PARD3 (Red) and EMA (Grey). (E) Quinacrine (Male chromosome specific stain, Red) and DAPI (Green) staining of male vs female fetal tail samples (F) Feature plot and UMAP plot depicting Xist expression within E10.5 and E11.5 gonad followed by bioinformatic segregation of female gonadal germ cells (XX only) to avoid Quinacrine false-negative by segregating the cells expressing significant amount of XX specific genes (eg: Xist, Ddx3x, Utx) depicted in UMAP plot as XX only cells. (G) Cluster of germ cells from E12, E14, E16, E18, P1 and P5 of previously published data (NCBI: Niu and Spradling, 2020) were used. E10, E11 and E15 (10X genomics) ScRNA seq was performed and germ cells isolated bioinformatically to create E10.5-P5 merged germ cell data to comprehensively look at gene expression across developmental stages. The number of cells used to create merged dataset is shown. (H) Feature plot validating all the cells express germ cell specific markers (Dppa3/DDx4) (I) Pre-meiotic/Meiotic gene expression analysis depicted by feature plot for various developmental stages within merged dataset. (J) Stack violin plot depicting Golgi-UPR pathway associated gene expression pattern across E10.5 to P5 meiotic germ cell and nurse cell developmental stages. Scale bar= 20 μ m (A,B-B, D) and 10 μ m (C), 100 μ m (E).

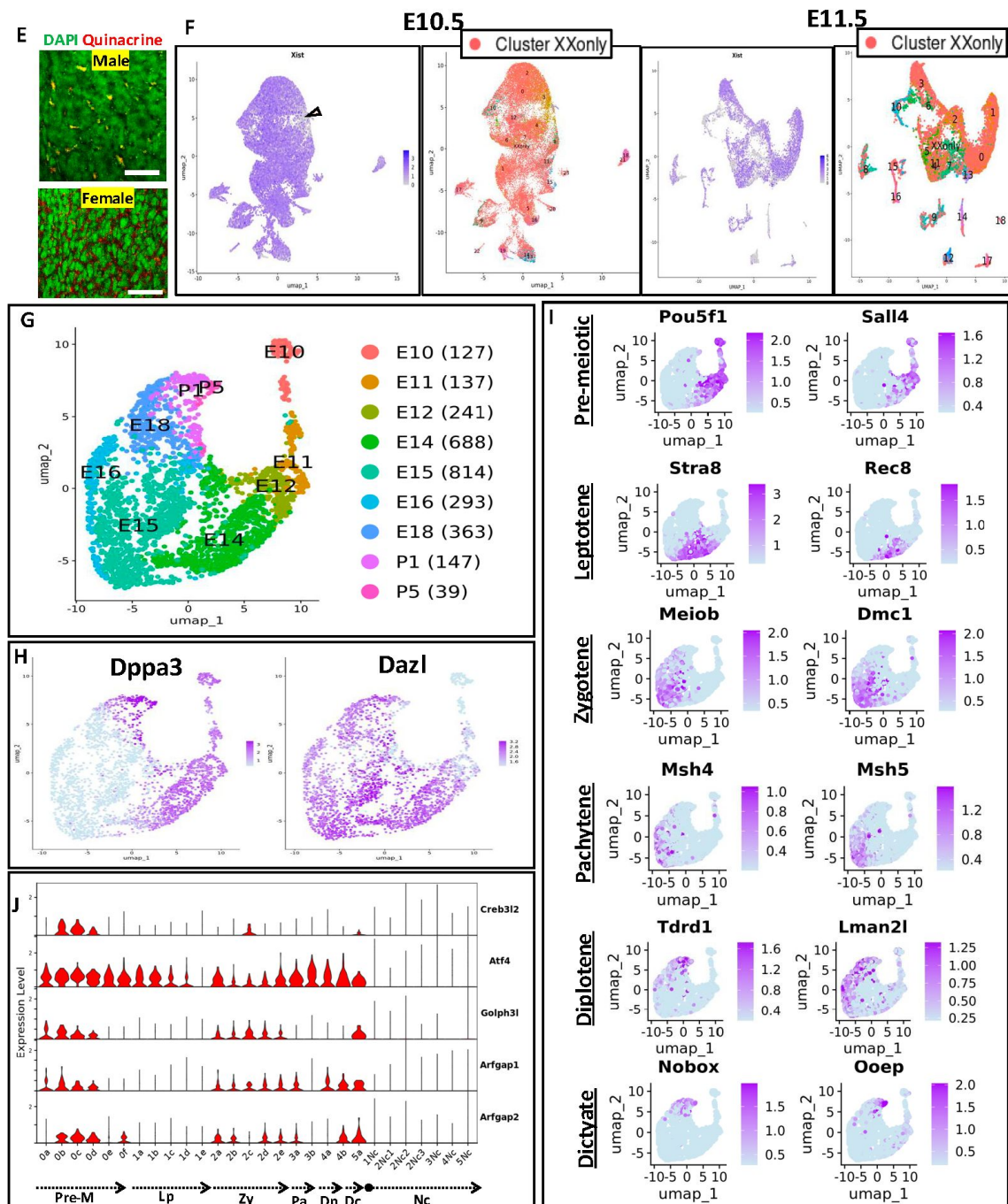


Fig. S3. (continued)

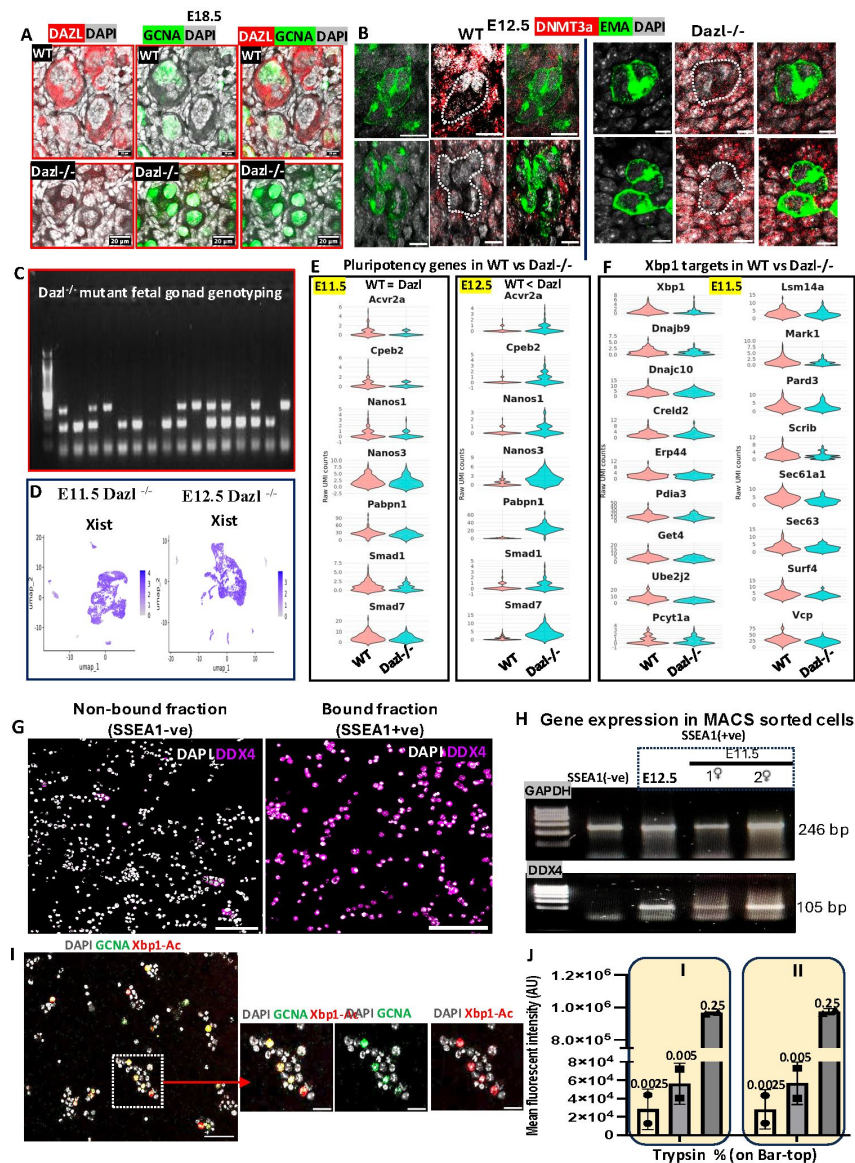


Fig. S4.

Validation of female *Dazl* mutant phenotype, MACS and Activity assays.

(A) Expected absence of *Dazl* protein in *Dazl* mutant gonad is shown by staining of E18.5 WT and *Dazl*^{-/-} gonad for *Dazl* (Red), GCNA (Green) and DAPI (Grey). (B) E12.5 WT and *Dazl*^{-/-} gonad stained for DNMT3a (Red), EMA (Green) and DAPI (Grey) (C) Fetal tail genotyping for ScRNA sequencing: Gel electrophoresis showing the standard *Dazl* genotyping PCR assay by Jackson (Stock No: 035880 Protocol 40585) with expected results: Wild type (192 bp), Heterozygous mutant (300 and 192 bp) and Homozygous mutant (300 bp). (D) Feature plot depicting positive *Xist* Expression in ScRNA seq data of E11.5 and E12.5 *Dazl* mutant fetal gonad, thus concluding them as female samples. (E) Stackviolin Plot depicting pluripotency gene expression pattern in E11.5-E12.5 WT vs *Dazl*^{-/-} gonad. (F) Stackviolin Plot depicting *Xbp1* targets gene expression pattern in E1.5 WT vs *Dazl*^{-/-} gonad (G) Staining of both unbound SSEA1(-ve) and the bound fraction consisting of SSEA1(+ve) cells for DDX4 (Magenta) and DAPI (Grey). (H) Gel electrophoresis showing the PCR amplification of housekeeping gene GAPDH and Germ cell specific marker DDX4 in SSEA1(-ve) and SSEA1(+ve) germ cells from E11.5 and E12.5 ovary. The amplicon size is indicated on left. (I) Proof of functioning of *Xbp1* assay is shown by staining of mixture of SSEA1(-ve) and SSEA1(+ve) cells for GCNA (Green), DAPI (Grey) and *Xbp1*(Red) (J) Proof of function of proteasome activity assay using Trypsin as technical positive control. The proteasome activity assay was shown for two different assays (I and II) with technical duplicates at three different Trypsin concentrations. Mean fluorescent intensity in arbitrary units is shown obtained using plate reader. Scale bar: 20 μ m (A), 10 μ m (B), 100 μ m (G and I).

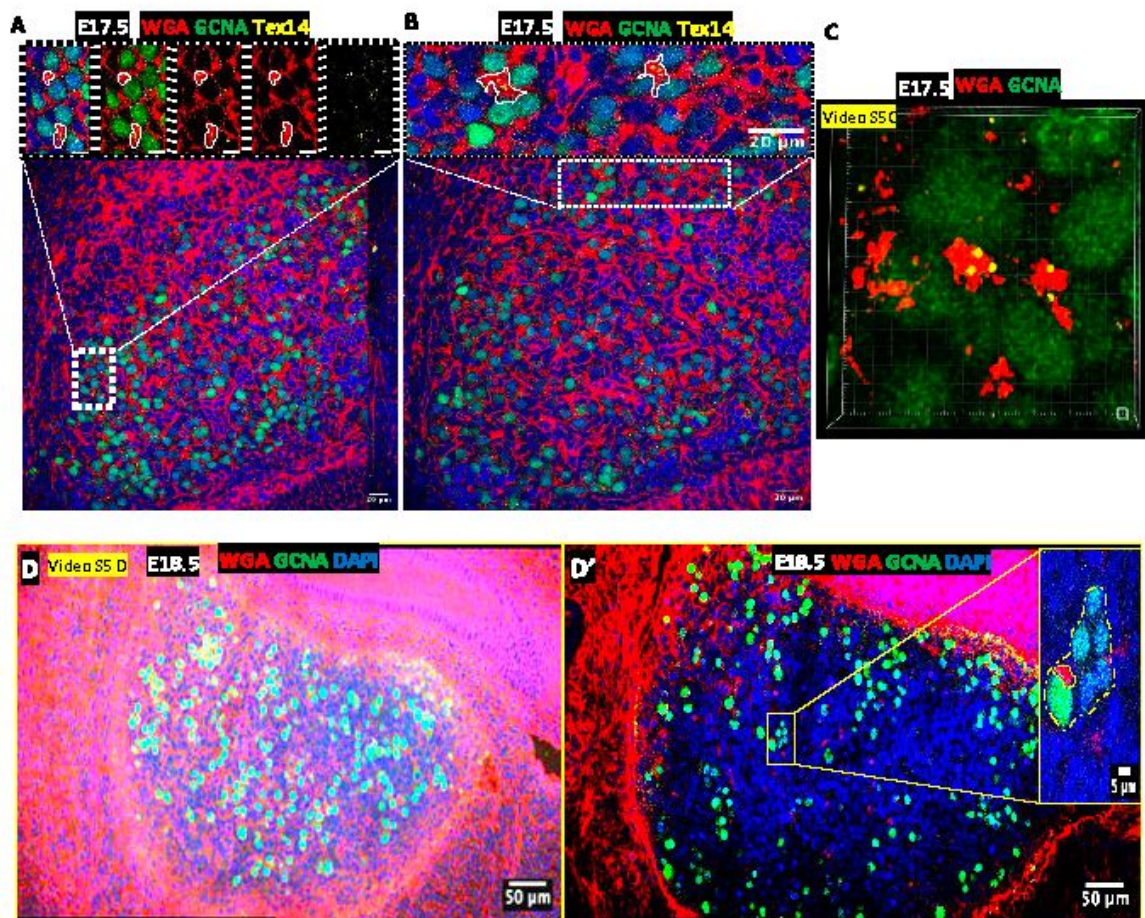


Fig. S5.

Visham enrichment within destined medullary Oocytes in WT vs *Dazl* mutant.

(A-B) Zoomed out E17.5 ovary covering large span of tissue stained with DAPI (Blue), WGA (Red), GCNA (Green) and Tex14 (Yellow). White dotted line is zoomed in and shown separately in main figure (Refer to main **Fig 5A and A'**). (C) Video created using Imari software showing E17.5 ovary stained for GCNA (Green), Tex14 (Yellow) and WGA (Red) followed by surface rendering of WGA aggregate (continuous one) associated with ring canals. (See Video S5C) (D) Video of E18.5 female gonad stained with DAPI (Blue), GCNA (Green) and WGA (Red) depicts the cortex and medullary region where medullary region showing distinct enriched WGA aggregate compared to surrounding small nurse cells. (See Video S5D) (D') One of the z-stack from E18.5 ovary Video in E stained with DAPI, GCNA and WGA depicts the medullary big Oocyte-like cells showing distinct enriched WGA aggregate compared to surrounding small nurse cells. E) Z-stack video of E18.5 WT, *Dazl* +/- and *Dazl* -/- gonad stained for GCNA (Green) and WGA (Red). (See Videos S5E1 to S5E3). (F) Z-stack video of P0 WT, *Dazl* +/- and *Dazl* -/- gonad stained for GCNA (Green) and PARD3 (Red). (See S5 F1 to F3) (G) E18.5 WT and *Dazl* +/- gonad stained for Mitotracker (Red) and GCNA (Green). Scale bar: 20 μ m (A-B and G), 50 μ m (D-E), 100 μ m (F).

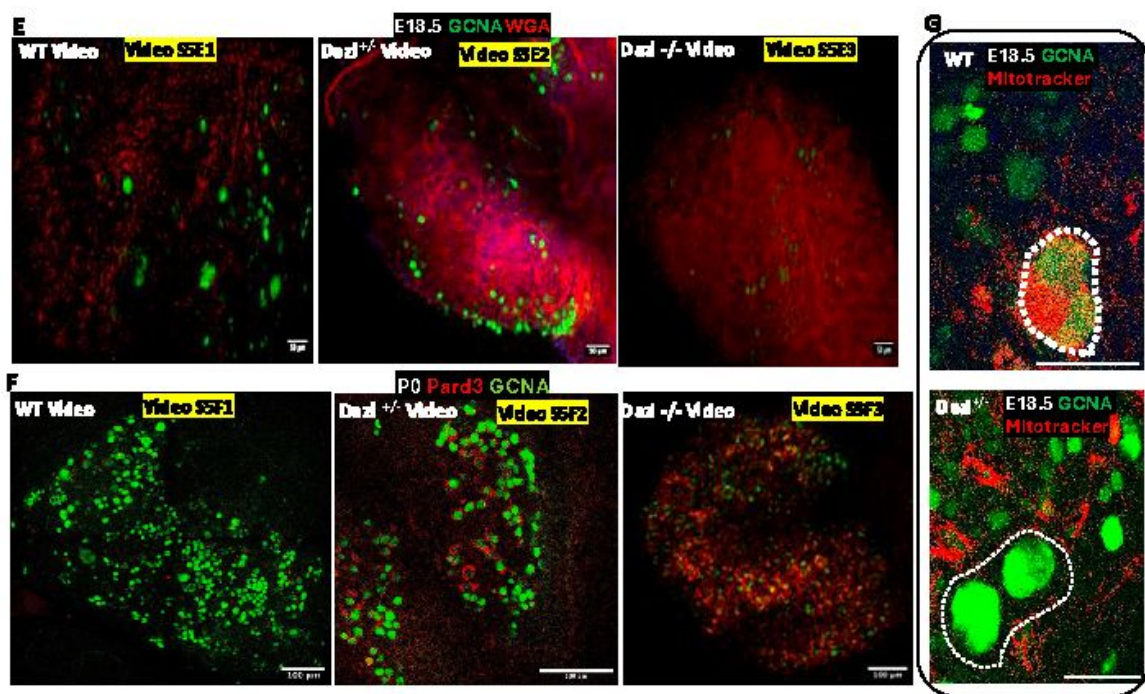


Fig. S5. (continued)

Additional information

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References

Apostolopoulos V., Stojanovska L., Gargosky S. E (2015) **MUC1 (CD227): A multi-tasked molecule** *Cellular and Molecular Life Sciences* **72**:4475–4500 <https://doi.org/10.1007/S00018-015-2014Z> | [Google Scholar](#)

Bohnert KA, Kenyon C (2017) **A lysosomal switch triggers proteostasis renewal in the immortal *C. elegans* germ lineage** *Nature* **551**:629–633 <https://doi.org/10.1038/nature24620> | [Google Scholar](#)

Anderson E., Beams H. W (1960) **Cytological observations on the fine structure of the guinea pig ovary with special reference to the oogonium, primary oocyte and associated follicle cells** *Journal of Ultrastructure Research* **3**:432–446 [https://doi.org/10.1016/S0022-5320\(60\)90021-6](https://doi.org/10.1016/S0022-5320(60)90021-6) | [Google Scholar](#)

Bertho S., Clapp M., Banisch T. U., Bandemer J., Raz E., Marlow F. L (2021) **Zebrafish *dazl* regulates cystogenesis and germline stem cell specification during the primordial germ cell to germline stem cell transition** *Development* **148** <https://doi.org/10.1242/DEV.187773> | [Google Scholar](#)

Büning J (1994) **The Insect Ovary: Ultrastructure, Previtellogenic Growth and Evolution** Chapman and Hall <https://doi.org/10.1007/978-94-011-0741-9> | [Google Scholar](#)

Chen H. H., Welling M., Bloch D. B., Muñoz J., Mientjes E., Chen X., Tramp C., Wu J., Yabuuchi A., Chou Y. F., Buecker C., Krainer A., Willemsen R., Heck A. J., Geijsen N (2014) **DAZL Limits Pluripotency, Differentiation, and Apoptosis in Developing Primordial Germ Cells** *Stem Cell Reports* **3**:892 <https://doi.org/10.1016/j.STEMCR.2014.09.003> | [Google Scholar](#)

Clark J. M., Eddy E. M (1975) **Fine structural observations on the origin and associations of primordial germ cells of the mouse** *Developmental Biology* **47**:136–155 [https://doi.org/10.1016/0012-1606\(75\)90269-9](https://doi.org/10.1016/0012-1606(75)90269-9) | [Google Scholar](#)

Conti M, Kunitomi C (2024) **A genome-wide perspective of the maternal mRNA translation program during oocyte development** *Semin Cell Dev Biol* **154**:88–98 <https://doi.org/10.1016/j.semcdb.2023.03.003> | [Google Scholar](#)

Cox D. N., Seyfried S. A., Jan L. Y., Jan Y. N (2001) **Bazooka and atypical protein kinase C are required to regulate oocyte differentiation in the *Drosophila* ovary** *Proceedings of the National Academy of Sciences of the United States of America* **98**:14475–14480 <https://doi.org/10.1073/pnas.261565198> | [Google Scholar](#)

Cox R. T., Spradling A. C (2003) **A Balbiani body and the fusome mediate mitochondrial inheritance during *Drosophila* oogenesis** *Development* **130**:1579–1590 <https://doi.org/10.1242/DEV.00365> | [Google Scholar](#)

- Cox R. T., Spradling A. C (2006) **Milton controls the early acquisition of mitochondria by *Drosophila* oocytes** *Development* **133**:3371–3377 <https://doi.org/10.1242/DEV.02514> | Google Scholar
- de Cuevas M., Spradling A. C. (1998) **Morphogenesis of the *Drosophila* fusome and its implications fo oocyte specification** *Development* **125**:2781–2789 <https://doi.org/10.1242/DEV.125.15.2781> | Google Scholar
- Fazel A. R., Schulte B. A., Spicer S. S (1990) **Glycoconjugate unique to migrating primordial germ cells differs with genera** *The Anatomical Record* **228**:177–184 <https://doi.org/10.1002/AR.1092280209> | Google Scholar
- Fazel A. R., Schulte B. A., Thompson R. P., Spicer S. S (1987) **Presence of a unique glycoconjugate on the surface of rat primordial germ cells during migration** *Cell Differentiation* **21**:199–211 [https://doi.org/10.1016/0045-6039\(87\)90456-8](https://doi.org/10.1016/0045-6039(87)90456-8) | Google Scholar
- Fichelson P., Huynh J. R (2007) **Asymmetric Divisions of Germline Cells** *Progress in Molecular and Subcellular Biology* **45**:97–120 https://doi.org/10.1007/978-3-540-69161-7_5 | Google Scholar
- Floros V. I., Pyle A., Dietmann S., Wei W., Tang W. W. C., Irie N., Payne B., Capalbo A., Noli L., Coxhead J., Hudson G., Crosier M., Strahl H., Khalaf Y., Saitou M., Ilic D., Surani M. A., Chinnery P. F. (2018) **Segregation of mitochondrial DNA heteroplasmy through a developmental genetic bottleneck in human embryos** *Nature Cell Biology* **2018**:144–151 <https://doi.org/10.1038/s41556-017-0017-8> | Google Scholar
- Fu X. F., Cheng S. F., Wang L. Q., Yin S., de Felici M., Shen W. (2015) **DAZ Family Proteins, Key Players for Germ Cell Development** *International Journal of Biological Sciences* **11**:1226 <https://doi.org/10.7150/IJBS.11536> | Google Scholar
- Giardina AG (1901) **Origine dell’oocyte e delle cellule nutritive nel *Dytiscus*** *Internationale Monatsschrift Fur Anatomie Und Physiologie* **18**:417–484 [Google Scholar](#)
- Gill M. E., Hu Y. C., Lin Y., Page D. C (2011) **Licensing of gametogenesis, dependent on RNA binding protein DAZL, as a gateway to sexual differentiation of fetal germ cells** *Proc. Natl. Acad. Sci* **108**:7443–7448 <https://doi.org/10.1073/pnas.1104501108> | Google Scholar
- Ginsburg M., Snow M. H. L., McLaren A (1990) **Primordial germ cells in the mouse embryo during gastrulation** *Development* **110**:521–528 <https://doi.org/10.1242/DEV.110.2.521> | Google Scholar
- Gondos B (1973) **Intercellular Bridges and Mammalian Germ Cell Differentiation** *Differentiation* **1**:177–182 <https://doi.org/10.1111/j.1432-0436.1973.TB00112.X> | Google Scholar
- Hahnel A. C., Eddy E. M (1987) **The distribution of two cell surface determinants of mouse embryonal carcinoma and early embryonic cells** *Journal of Reproductive Immunology* **10**:89–110 [https://doi.org/10.1016/0165-0378\(87\)90069-6](https://doi.org/10.1016/0165-0378(87)90069-6) | Google Scholar
- Haston K. M., Tung J. Y., Reijo Pera R. A (2009) **Dazl Functions in Maintenance of Pluripotency and Genetic and Epigenetic Programs of Differentiation in Mouse Primordial Germ Cells In Vivo and In Vitro** *PLOS One* **4**:e5654 <https://doi.org/10.1371/JOURNAL.PONE.0005654> | Google Scholar

- Heasman J., Quarmby J., Wylie C. C (1984) **The mitochondrial cloud of *Xenopus* oocytes: The source of germinal granule material** *Developmental Biology* **105**:458–469 [https://doi.org/10.1016/0012-1606\(84\)90303-8](https://doi.org/10.1016/0012-1606(84)90303-8) | [Google Scholar](#)
- Hegner R. W (1914) **Studies on germ cells. I. The history of the germ cells in insects with special reference to the Keimbahn determinants. II. The origin and significance of the Keimbahn determinants in animals** *Journal of Morphology* **25**:375–509 <https://doi.org/10.1002/JMOR.1050250302> | [Google Scholar](#)
- Hen G., Friedman-Einat M., Sela-Donenfeld D (2014) **Primordial germ cells in the dorsal mesentery of the chicken embryo demonstrate left-right asymmetry and polarized distribution of the EMA1 epitope** *Journal of Anatomy* **224**:556–563 <https://doi.org/10.1111/JOA.12163> | [Google Scholar](#)
- Hill J. H., Chen Z., Xu H (2014) **Selective propagation of functional mitochondrial DNA during oogenesis restricts the transmission of a deleterious mitochondrial variant** *Nat. Genet* **46**:389–392 [Google Scholar](#)
- Hinnant T. D., Merkle J. A., Ables E. T (2020) **Coordinating Proliferation, Polarity, and Cell Fate in the *Drosophila* Female Germline** *Frontiers in Cell and Developmental Biology* **8**:19 <https://doi.org/10.3389/FCELL.2020.00019> | [Google Scholar](#)
- Holliday R (1984) **The biological significance of meiosis** *Symp. Soc. Exp. Biol* :38381–394 [Google Scholar](#)
- Huynh J.-R., Petronczki M., Knoblich J.A., St Johnston D (2001) **Bazooka and PAR-6 are required with PAR-1 for the maintenance of oocyte fate in *Drosophila*** *Curr. Biol* **11**:901–906 [Google Scholar](#)
- Huynh J. R., Shulman J. M., Benton R., St Johnston D (2001) **PAR-1 is required for the maintenance of oocyte fate in *Drosophila*** *Development* **128**:1201–1209 <https://doi.org/10.1242/DEV.128.7.1201> | [Google Scholar](#)
- Huynh JR, St Johnston D (2004) **The origin of asymmetry: early polarisation of the *Drosophila* germline cyst and oocyte** *Curr Biol* **14**:R438–49 <https://doi.org/10.1016/j.cub.2004.05.040> | [Google Scholar](#)
- Ikami K., Shoffner-Beck S., Weh M. T., Schnell S., Yoshida S., Miranda E. A. D., Ko S., Lei L (2023) **Branched germline cysts and female-specific cyst fragmentation facilitate oocyte determination in mice** *Proceedings of the National Academy of Sciences of the United States of America* **120** <https://doi.org/10.1073/PNAS.2219683120> | [Google Scholar](#)
- Jeon K. W., Kennedy J. R (1973) **The primordial germ cells in early mouse embryos: Light and electron microscopic studies** *Developmental Biology* **31**:275–284 [https://doi.org/10.1016/0012-1606\(73\)90264-9](https://doi.org/10.1016/0012-1606(73)90264-9) | [Google Scholar](#)
- Jung J. G., Kim D. K., Park T. S., Lee S. D., Lim J. M., Han J. Y (2005) **Development of Novel Markers for the Characterization of Chicken Primordial Germ Cells** *Stem Cells* **23**:689–698 <https://doi.org/10.1634/STEMCELLS.2004-0208> | [Google Scholar](#)
- Kaltschmidt J. A., Brand A. H (2002) **Asymmetric cell division: microtubule dynamics and spindle asymmetry** *Journal of Cell Science* **115**:2257–2264 <https://doi.org/10.1242/JCS.115.11.2257> | [Google Scholar](#)

- Kemphues K. J., Priess J. R., Morton D. G., Cheng N (1988) **Identification of genes required for cytoplasmic localization in early C. elegans embryos** *Cell* **52**:311–320 [https://doi.org/10.1016/S0092-8674\(88\)80024-2](https://doi.org/10.1016/S0092-8674(88)80024-2) | [Google Scholar](#)
- King RC (1970) **Oogenesis in Drosophila melanogaster** New York: Academic Press [Google Scholar](#)
- Kloc M, Bilinski S, Dougherty MT, Brey EM, Etkin LD (2004a) **Formation, architecture and polarity of female germline cyst in Xenopus** *Developmental Biology* **266**:43–61 <https://doi.org/10.1016/j.ydbio.2003.10.002> | [Google Scholar](#)
- Kloc M., Bilinski S., Etkin L. D (2004b) **The Balbiani Body and Germ Cell Determinants: 150 Years Later** *Current Topics in Developmental Biology* **59**:1–36 [https://doi.org/10.1016/S0070-2153\(04\)59001-4](https://doi.org/10.1016/S0070-2153(04)59001-4) | [Google Scholar](#)
- Lawson K. A., Dunn N. R., Roelen B. A. J., Zeinstra L. M., Davis A. M., Wright C. V. E., Korving J. P. W. F. M., Hogan B. L. M (1999) **Bmp4 is required for the generation of primordial germ cells in the mouse embryo** *Genes and Development* **13**:424–436 <https://doi.org/10.1101/GAD.13.4.424> | [Google Scholar](#)
- Lei L, Spradling AC (2013) **Female mice lack adult germ-line stem cells but sustain oogenesis using stable primordial follicles** *Proc. Natl. Acad. Sci* **110**:8585–8590 <https://doi.org/10.1073/PNAS.1306189110/-/DCSUPPLEMENTAL> | [Google Scholar](#)
- Lei L., Spradling A. C (2016) **Mouse oocytes differentiate through organelle enrichment from sister cyst germ cells** *Science* **352**:95 <https://doi.org/10.1126/SCIENCE.AAD215> | [Google Scholar](#)
- Levy EW, Leite I, Joyce BW, Shvartsman SY, Posfai E (2024) **A tug-of-war between germ cell motility and intercellular bridges controls germline cyst formation in mice** *Current Biology* **34**:5728–5738 <https://doi.org/10.1016/j.cub.2024.10.062> | [Google Scholar](#)
- Lieber T., Jeedigunta S. P., Palozzi J. M., Lehmann R., Hurd T. R (2019) **Mitochondrial fragmentation drives selective removal of deleterious mtDNA in the germline** *Nature* **570**:380–384 <https://doi.org/10.1038/s41586-019-1213-4> | [Google Scholar](#)
- Lighthouse D, Buszczak M, Spradling AC (2008) **New components of the Drosophila fusome suggest it plays novel roles in signaling and transport** *Developmental Biology* **317**:59–71 <https://doi.org/10.1016/j.YDBIO.2008.02.009> | [Google Scholar](#)
- Lin H, Yue L, Spradling AC (1994) **The Drosophila fusome, a germline-specific organelle, contains membrane skeletal proteins and functions in cyst formation** *Development* **120**:947–956 <https://doi.org/10.1242/DEV.120.4.947> | [Google Scholar](#)
- Littlefield CL (1994) **Cell-cell interactions and the control of sex determination in hydra.** *Semin Dev. Biol* **5**:13–20 <https://doi.org/10.1006/sedb.1994.1003> | [Google Scholar](#)
- Liu Y., Lau X., Munusamy P., Sanchez C. M. A. S., Snell D., Sangrithi M (2025) **Single-cell RNA-seq identifies protracted mouse germline X chromosome reactivation dynamics directed by a PRC2-dependent mechanism** *Developmental Cell* **60**:1321–1335 <https://doi.org/10.1016/j.DEVCEL.2024.12.028/ASSET/714FBA10-362A-4CC7-AA61-56D56DC3A471/MAIN.ASSETS/GR4.JPG> | [Google Scholar](#)

- Loda A., Collombet S., Heard E (2022) **Gene regulation in time and space during X-chromosome inactivation** *Nature Reviews Molecular Cell Biology* **23**:231–249 <https://doi.org/10.1038/s41580-021-00438-7> | [Google Scholar](#)
- Machamer C. E (2015) **The Golgi complex in stress and death** *Frontiers in Neuroscience* **9** <https://doi.org/10.3389/FNINS.2015.00421> | [Google Scholar](#)
- Mahowald A. P., Strassheim J. M (1970) **Intercellular migration of centrioles in the germarium of *Drosophila Melanogaster*: An electron microscopic study** *Journal of Cell Biology* **45**:306–320 <https://doi.org/10.1083/JCB.45.2.306> | [Google Scholar](#)
- Marlow F. L., Mullins M. C (2008) **Bucky ball functions in Balbiani body assembly and animal-vegetal polarity in the oocyte and follicle cell layer in zebrafish** *Developmental Biology* **321**:40–50 <https://doi.org/10.1016/j.ydbio.2008.05.557> | [Google Scholar](#)
- Matova N, Cooley L (2001) **Comparative Aspects of Animal Oogenesis** *Developmental Biology* **231**:291–320 <https://doi.org/10.1006/DBIO.2000.0120> | [Google Scholar](#)
- Matsui T, Watanabe T, Matsuzawa K, Kakeno M, Okumura N, Sugiyama I, Itoh N, Kaibuchi K. (2015) **PAR3 and aPKC regulate Golgi organization through CLASP2 phosphorylation to generate cell polarity.** *Molecular Biology of the Cell* **26**:751 <https://doi.org/10.1091/MBC.E14-09-1382> | [Google Scholar](#)
- Meiring J. C. M., Shneyer B. I., Akhmanova A (2020) **Generation and regulation of microtubule network asymmetry to drive cell polarity** *Current Opinion in Cell Biology* **62**:86–95 <https://doi.org/10.1016/J.CEB.2019.10.004> | [Google Scholar](#)
- Mercer M., Jang S., Ni C., Buszczak M (2021) **The Dynamic Regulation of mRNA Translation and Ribosome Biogenesis During Germ Cell Development and Reproductive Aging** *Frontiers in Cell and Developmental Biology* **9** <https://doi.org/10.3389/FCELL.2021.710186> | [Google Scholar](#)
- Monteiro VL, Safavian D, Vasudevan D, Hurd TR (2023) **Mitochondrial remodelling is essential for female germ cell differentiation and survival** *PLoS Genet* **19**:e1010610 <https://doi.org/10.1371/journal.pgen.1010610> | [Google Scholar](#)
- Moore C. A., Zernicka-Goetz M (2005) **PAR-1 and the microtubule-associated proteins CLASP2 and dynactin-p50 have specific localisation on mouse meiotic and first mitotic spindles** *Reproduction* **130**:311–320 <https://doi.org/10.1530/REP.1.00651> | [Google Scholar](#)
- Nicholls P. K., Schorle H., Naqvi S., Hu Y. C., Fan Y., Carmell M. A., Dobrinski I., Watson A. L., Carlson D. F., Fahrenkrug S. C., Page D. C (2019) **Mammalian germ cells are determined after PGC colonization of the nascent gonad** *Proc. Natl. Acad. Sci. USA* **116**:25677–25687 <https://doi.org/10.1073/pnas.1910733116> | [Google Scholar](#)
- Niu W., Spradling A. C (2020) **Two distinct pathways of pregranulosa cell differentiation support follicle formation in the mouse ovary** *Proc. Natl. Acad. Sci. USA* **117**:20015–20026 <https://doi.org/10.1073/PNAS.2005570117> | [Google Scholar](#)
- Niu W., Spradling A. C (2022) **Mouse oocytes develop in cysts with the help of nurse cells** *Cell* **185**:2576–2590 <https://doi.org/10.1016/J.CELL.2022.05.001> | [Google Scholar](#)
- Palozzi J. M., Jeedigunta S. P., Minenkova A. V., Monteiro V. L., Thompson Z. S., Lieber T., Hurd T. R. (2022) **Mitochondrial DNA quality control in the female germline requires a unique**

programmed mitophagy *Cell Metabolism* **34**:1809–1823 <https://doi.org/10.1016/j.cmet.2022.10.005> | [Google Scholar](#)

Pang L. Y., DeLuca S., Zhu H., Urban J. M., Spradling A. C (2023) **Chromatin and gene expression changes during female *Drosophila* germline stem cell development illuminate the biology of highly potent stem cells** *eLife* **12** <https://doi.org/10.7554/ELIFE.90509> | [Google Scholar](#)

Pepling M. E., de Cuevas M., Spradling A. C. (1999) **Germline cysts: a conserved phase of germ cell development?** *Trends in Cell Biology* **9**:257–262 [https://doi.org/10.1016/S0962-8924\(99\)01594-9](https://doi.org/10.1016/S0962-8924(99)01594-9) | [Google Scholar](#)

Pepling M. E., Spradling A. C (1998) **Female mouse germ cells form synchronously dividing cysts** *Development* **125**:3323–3328 <https://doi.org/10.1242/DEV.125.17.3323> | [Google Scholar](#)

Pepling M. E., Spradling A. C (2001) **Mouse ovarian germ cell cysts undergo programmed breakdown to form primordial follicles** *Dev. Biol* **234**:339–351 <https://doi.org/10.1006/DBIO.2001.0269> | [Google Scholar](#)

Pepling M. E., Wilhelm J. E., O'Hara A. L., Gephardt G. W., Spradling A. C (2007) **Mouse oocytes within germ cell cysts and primordial follicles contain a Balbiani body** *PNAS* **104**:187–192 <https://doi.org/10.1073/pnas.0609923104> | [Google Scholar](#)

Petronczki M., Knoblich J. A (2001) **DmPAR-6 directs epithelial polarity and asymmetric cell division of neuroblasts in *Drosophila*** *Nature Cell Biol* **3**:43–49 <https://doi.org/10.1038/35050550> | [Google Scholar](#)

Planelles-Herrero V. J., Bittleston A., Seum C., Gaitan M. G., Derivery E (2022) **Elongator stabilizes microtubules to control central spindle asymmetry and polarized trafficking of cell fate determinants** *Nature Cell Biol* **24**:1606–1616 <https://doi.org/10.1038/s41556-022-01020-9> | [Google Scholar](#)

Reik W., Azim Surani M (2015) **Germline and Pluripotent Stem Cells** *Cold Spring Harbor Perspectives in Biology* **7**:a019422 <https://doi.org/10.1101/CSHPERSPECT.A019422> | [Google Scholar](#)

Saitou M., Kagiwada S., Kurimoto K (2011) **Epigenetic reprogramming in mouse pre-implantation development and primordial germ cells** *Development* **139**:15–31 <https://doi.org/10.1242/DEV.050849> | [Google Scholar](#)

Seki Y., Yamaji M., Yabuta Y., Sano M., Shigeta M., Matsui Y., Saga Y., Tachibana M., Shinkai Y., Saitou M (2007) **Cellular dynamics associated with the genome-wide epigenetic reprogramming in migrating primordial germ cells in mice** *Development* **134**:2627–2638 <https://doi.org/10.1242/dev.005611> | [Google Scholar](#)

Suda Y., Tachikawa H., Suda T., Kurokawa K., Nakano A., Irie K (2024) **Remodeling of the secretory pathway is coordinated with de novo membrane formation in budding yeast gametogenesis** *IScience* **27** <https://doi.org/10.1016/j.isci.2024.110855> | [Google Scholar](#)

Sekula M., Tworzydło W., Bilinski S. M (2024) **Balbani body of basal insects is potentially involved in multiplication and selective elimination of mitochondria** *Scientific Reports* **14** <https://doi.org/10.1038/S41598-024-58997-6> | [Google Scholar](#)

- Snapp E. L., Iida T., Frescas D., Lippincott-Schwartz J., Lilly M. A (2004) **The Fusome Mediates Intercellular Endoplasmic Reticulum Connectivity in Drosophila Ovarian Cysts** *Molecular Biology of the Cell* **15**:4512 <https://doi.org/10.1091/MBC.E04-06-0475> | [Google Scholar](#)
- Spradling A. C., Niu W., Yin Q., Pathak M., Maurya B (2022) **Conservation of oocyte development in germline cysts from Drosophila to mouse** *eLife* **11** <https://doi.org/10.7554/ELIFE.83230> | [Google Scholar](#)
- Spradling A.C (2023) **The ancient origin and function of germline cysts** *Results Probl Cell Differ* **71**:3–21 <https://doi.org/10.1007/978-3-031-37936-9> | [Google Scholar](#)
- Sunchu B., Cabernard C (2020) **Principles and mechanisms of asymmetric cell division** *Development* **147**:dev167650 <https://doi.org/10.1242/DEV.167650> | [Google Scholar](#)
- Unal E, Kinde B, Amon A (2011) **Gametogenesis eliminates age-induced cellular damage and resets life span in yeast** *Science* **332**:1554–1557 [Google Scholar](#)
- Urven L. E., Erickson C. A., Abbott U. K., McCarrey J. R (1988) **Analysis of germ line development in the chick embryo using an antimouse EC cell antibody** *Development* **103**:299–304 <https://doi.org/10.1242/DEV.103.2.299> | [Google Scholar](#)
- Vaccari T., Ephrussi A (2002) **The fusome and microtubules enrich Par-1 in the oocyte, where it effects polarization in conjunction with Par-3, BicD, Egl, and Dynein** *Current Biology* **12**:1524–1528 [https://doi.org/10.1016/S0960-9822\(02\)01079-5](https://doi.org/10.1016/S0960-9822(02)01079-5) | [Google Scholar](#)
- Vembar S., Brodsky J (2008) **One step at a time: endoplasmic reticulum-associated degradation** *Nat Rev Mol Cel Biol* **9**:944–957 <https://doi.org/10.1038/nrm2546> | [Google Scholar](#)
- Watson J. L., Krüger L. K., Ben-Sasson A. J., Bittleston A., Shahbazi M. N., Planelles-Herrero V. J., Chambers J. E., Manton J. D., Baker D., Derivery E (2023) **Synthetic Par polarity induces cytoskeleton asymmetry in unpolarized mammalian cells** *Cell* **186**:4710–4727 <https://doi.org/10.1016/J.CELL.2023.08.034> | [Google Scholar](#)
- Weismann A (1892) **Das Keimplasma** Jena: G. Fisher [Google Scholar](#)
- Yamashita Y. M (2023) **Asymmetric Stem Cell Division and Germline Immortality** *Annual Review of Genetics* **57**:181–199 <https://doi.org/10.1146/ANNUREV-GENET-022123-040039> | [Google Scholar](#)
- Yang CR., Rajkovic G., Daldello E.M., et al. (2020) **The RNA-binding protein DAZL functions as repressor and activator of mRNA translation during oocyte maturation** *Nat Commun* **11**:1399 <https://doi.org/10.1038/s41467-020-15209-9> | [Google Scholar](#)
- Yin Q., Spradling A. C. (2025) **Distinct waves of ovarian follicles contribute to mouse oocyte production** *BioRxiv* :2024.05.30.596686 <https://doi.org/10.1101/2024.05.30.596686> | [Google Scholar](#)
- Yoshinaga K., Fujimoto T., Nakamura M., Terakura H (1992) **Selective lectin-binding sites of primordial germ cells in chick and quail embryos** *The Anatomical Record* **233**:625–632 <https://doi.org/10.1002/AR.1092330416> | [Google Scholar](#)
- Yoshinaga K., Muramatsu H., Muramatsu T (1991) **Immunohistochemical localization of the carbohydrate antigen 4C9 in the mouse embryo: a reliable marker of mouse primordial**

germ cells *Differentiation* **48**:75–82 <https://doi.org/10.1111/j.1432-0436.1991.tb00245.x> | [Google Scholar](#)

Xiao T, Ünal E (2025) **Remodeling, compartmentalization, and degradation: a trifecta for organelle quality control during gametogenesis** *Curr Opin Genet Dev* **92**:102347 <https://doi.org/10.1016/j.gde.2025.102347> | [Google Scholar](#)

Zaffagnini G, Cheng S, Salzer MC, Pernaute B, Duran JM, Irimia M, Schuh M, Böke E (2024) **Mouse oocytes sequester aggregated proteins in degradative super-organelles** *Cell* **2024**:1109–1126 [Google Scholar](#)

Zagore L. L., Sweet T. J., Hannigan M. M., Weyn-Vanhentenryck S. M., Jobava R., Hatzoglou M., Zhang C., Licatalosi D. D (2018) **DAZL Regulates Germ Cell Survival through a Network of PolyA-Proximal mRNA Interactions** *Cell Reports* **25**:1225–1240 <https://doi.org/10.1016/j.celrep.2018.10.012> | [Google Scholar](#)

McFarlane L., Truong V., Palmer J. S., Wilhelm D (2013) **Novel PCR Assay for Determining the Genetic Sex of Mice** *Sexual Development* **7**:207–211 <https://doi.org/10.1159/000348677> | [Google Scholar](#)

Gao L., Kim Y., Kim B., Lofgren S. M., Schultz-Norton J. R., Nardulli A. M., Heckert L. L., Jorgensen J. S (2011) **Two Regions Within the Proximal Steroidogenic Factor 1 Promoter Drive Somatic Cell-Specific Activity in Developing Gonads of the Female Mouse** *Biology of Reproduction* **84**:422–434 <https://doi.org/10.1095/BIOLREPROD.110.084590> | [Google Scholar](#)

Chunmei W., Huang C., Gu Y., Zhou Y., Zhu Z., Zhang Y (2011) **Biosynthesis and distribution of Lewis X-and Lewis Y-containing glycoproteins in the murine male reproductive system** *Glycobiology* **21**:225–234 <https://doi.org/10.1093/GLYCOB/CWQ152> | [Google Scholar](#)

Pathak M, Spradling AC (2025) **Mouse germline cysts contain a fusome that mediates oocyte development** *NCBI Gene Expression Omnibus* ID GSE303512 <http://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE303512>

Niu W, Spradling AC (2022) **Single-cell analysis RNA sequencing of prenatal and neonatal gonads/ovaries at E11.5, E12.5, E14.5, E16.5, E18.5, P1 and P5** *NCBI Gene Expression Omnibus* ID GSE136441 <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE136441>

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Reviewer #1 (Public review):

Summary:

The authors attempt to study how oocyte incomplete cytokinesis occurs in the mouse ovary.

Strengths:

The finding that UPR components are highly expressed during zygotene is an interesting result that has broad implications for how germ cells navigate meiosis. The findings that proteasome activity increases in germ cells compared to somatic cells suggest that the germline might have a quantitatively different response for protein clearance.

Weaknesses:

(1) The microscopy images look saturated, for example, Figure 1a, b, etc? Is this a normal way to present fluorescent microscopy?

(2) The authors should ensure that all claims regarding enrichment/lower vs lower values have indicated statistical tests.

(a) In Figure 2f, the authors should indicate which comparison is made for this test. Is it comparing 2 vs 6 cyst numbers?

(b) Figures 4d and 4e do not have a statistical test indicated.

(3) Because the system is developmentally dynamic, the major conclusions of the work are somewhat unclear. Could the authors be more explicit about these and enumerate them more clearly in the abstract?

(4) The references for specific prior literature are mostly missing (lines 184-195, for example).

(5) The authors should define all acronyms when they are first used in the text (UPR, EGAD, etc).

(6) The jumping between topics (EMA, into microtubule fragmentation, polarization proteins, UPR/ERAD/EGAD, GCNA, ER, balbiani body, etc) makes the narrative of the paper very difficult to follow.

(7) The heading title "Visham participates in organelle rejuvenation during meiosis" in line 241 is speculative and/or not supported. Drawing upon the extensive, highly rigorous *Drosophila* literature, it is safe to extrapolate, but the claim about regeneration is not adequately supported.

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Reviewer #2 (Public review):

This study identifies Visham, an asymmetric structure in developing mouse cysts resembling the *Drosophila* fusome, an organelle crucial for oocyte determination. Using immunofluorescence, electron microscopy, 3D reconstruction, and lineage labeling, the authors show that primordial germ cells (PGCs) and cysts, but not somatic cells, contain an EMA-rich, branching structure that they named Visham, which remains unbranched in male cysts. Visham accumulates in regions enriched in intercellular bridges, forming clusters reminiscent of fusome "rosettes." It is enriched in Golgi and endosomal vesicles and partially overlaps with the ER. During cell division, Visham localizes near centrosomes in interphase and early metaphase, disperses during metaphase, and reassembles at spindle poles during telophase before becoming asymmetric. Microtubule depolymerization disrupts its formation.

Cyst fragmentation is shown to be non-random, correlating with microtubule gaps. The authors propose that 8-cell (or larger) cysts fragment into 6-cell and 2-cell cysts. Analysis of *Pard3* (the mouse ortholog of *Par3/Baz*) reveals its colocalization with Visham during cyst asymmetry, suggesting that mammalian oocyte polarization depends on a conserved system involving *Par* genes, cyst formation, and a fusome-like structure.

Transcriptomic profiling identifies genes linked to pluripotency and the unfolded protein response (UPR) during cyst formation and meiosis, supported by protein-level reporters monitoring *Xbp1* splicing and 20S proteasome activity. Visham persists in meiotic germ cells at stage E17.5 and is later transferred to the oocyte at E18.5 along with mitochondria and Golgi vesicles, implicating it in organelle rejuvenation. In *Dazl* mutants, cysts form, but Visham dynamics, polarity, rejuvenation, and oocyte production are disrupted, highlighting its potential role in germ cell development.

Overall, this is an interesting and comprehensive study of a conserved structure in the germline cells of both invertebrate and vertebrate species. Investigating these early stages of germ cell development in mice is particularly challenging. Although primarily descriptive, the study represents a remarkable technical achievement. The images are generally convincing, with only a few exceptions.

Major comments:

(1) Some titles contain strong terms that do not fully match the conclusions of the corresponding sections.

(1a) Article title "Mouse germline cysts contain a fusome-like structure that mediates oocyte development":

The term "mediates" could be misleading, as the functional data on Visham (based on comparing its absence to wild-type) actually reflects either a microtubule defect or a *Dazl* mutant context. There is no specific loss-of-function of *visham* only.

(1b) Result title, "Visham overlaps centrosomes and moves on microtubules":

The term "moves" implies dynamic behavior, which would require live imaging data that are not described in the article.

(1c) Result title, "Visham associates with Golgi genes involved in UPR beginning at the onset of cyst formation":

The presented data show that the presence of Visham in the cyst coincides temporally with the expression and activity of the UPR response; the term "associates" is unclear in this context.

(1d) Result title, "Visham participates in organelle rejuvenation during meiosis":

The term "participates" suggests that Visham is required for this process, whereas the conclusion is actually drawn from the *Dazl* mutant context, not a specific loss-of-function of visham only.

(2) The authors aim to demonstrate that Visham is a fusome-like structure. I would suggest simply referring to it as a "fusome-like structure" rather than introducing a new term, which may confuse readers and does not necessarily help the authors' goal of showing the conservation of this structure in *Drosophila* and *Xenopus* germ cells. Interestingly, in a preprint from the same laboratory describing a similar structure in *Xenopus* germ cells, the authors refer to it as a "fusome-like structure (FLS)" (Davidian and Spradling, *BioRxiv*, 2025).

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Reviewer #3 (Public review):

This manuscript provides evidence that mice have a fusome, a conserved structure most well studied in *Drosophila* that is important for oocyte specification. Overall, a myriad of evidence is presented demonstrating the existence of a mouse fusome that the authors term visham. This work is important as it addresses a long-standing question in the field of whether mice have fusomes and sheds light on how oocytes are specified in mammals. Concerns that need to be addressed revolve around several conclusions that are overstated or unclear and are listed below.

(1) Line 86 - the heading for this section is "PGCs contain a Golgi-rich structure known as the EMA granule" but there is nothing in this section that shows it is Golgi-rich. It does show that the structure is asymmetric and has branches.

(2) Line 105-106, how do we know if what's seen by EM corresponds to the EMA1 granule?

(3) Line 106-107-states "Visham co-stained with the Golgi protein Gm130 and the recycling endosomal protein Rab11a1". This is not convincing as there is only one example of each image, and both appear to be distorted.

(4) Line 132-133---while visham formation is disrupted when microtubules are disrupted, I am not convinced that visham moves on microtubules as stated in the heading of this section.

(5) Line 156 - the heading for this section states that Visham associates with polarity and microtubule genes, including *pard3*, but only evidence for *pard3* is presented.

(6) Lines 196-210 - it's strange to say that UPR genes depend on *DAZ*, as they are upregulated in the mutants. I think there are important observations here, but it's unclear what is being concluded.

(7) Line 257-259---wave 1 and 2 follicles need to be explained in the introduction, and how this fits with the observations here clarified.

<https://doi.org/10.7554/eLife.109358.1.sa1>

Author response:

Reviewer #1 (Public Review):

Summary

We thank the reviewer for the constructive and thoughtful evaluation of our work. We appreciate the recognition of the novelty and potential implications of our findings regarding UPR activation and proteasome activity in germ cells.

(1) The microscopy images look saturated, for example, Figure 1a, b, etc. Is this a normal way to present fluorescent microscopy?

The apparent saturation was not present in the original images, but likely arose from image compression during PDF generation. While the EMA granule was still apparent, in the revised submission, we will provide high-resolution TIFF files to ensure accurate representation of fluorescence intensity and will carefully optimize image display settings to avoid any saturation artifacts.

(2) The authors should ensure that all claims regarding enrichment/lower vs. lower values have indicated statistical tests.

We fully agree. In the revised version, we will correct any quantitative comparisons where statistical tests were not already indicated, with a clear statement of the statistical tests used, including p-values in figure legends and text.

(a) In Figure 2f, the authors should indicate which comparison is made for this test. Is it comparing 2 vs. 6 cyst numbers?

We acknowledge that the description was not sufficiently detailed. Indeed, the test was not between 2 vs 6 cyst numbers, but between all possible ways 8-cell cysts or the larger cysts studied could fragment randomly into two pieces, and produce by chance 6-cell cysts in 13 of 15 observed examples. We will expand the legend and main text to clarify that a binomial test was used to determine that the proportion of cysts producing 6-cell fragments differed very significantly from chance.

Revised text:

“A binomial test was used to assess whether the observed frequency of 6-cell cyst products differed from random cyst breakage. Production of 6-cell cysts was strongly preferred (13/15 cysts; **** $p < 0.0001$).”

(b) Figures 4d and 4e do not have a statistical test indicated.

We will include the specific statistical test used and report the corresponding p-values directly in the figure legends.

(3) Because the system is developmentally dynamic, the major conclusions of the work are somewhat unclear. Could the authors be more explicit about these and enumerate them more clearly in the abstract?

We will revise the abstract to better clarify the findings of this study. We will also replace the term Visham with mouse fusome to reflect its functional and structural analogy to the Drosophila and Xenopus fusomes, making the narrative more coherent and conclusive.

(4) The references for specific prior literature are mostly missing (lines 184-195, for example).

We appreciate this observation of a problem that occurred inadvertently when shortening an earlier version. We will add 3–4 relevant references to appropriately support this section.

(5) The authors should define all acronyms when they are first used in the text (UPR, EGAD, etc).

We will ensure that all acronyms are spelled out at first mention (e.g., Unfolded Protein Response (UPR), Endosome and Golgi-Associated Degradation (EGAD)).

(6) The jumping between topics (EMA, into microtubule fragmentation, polarization proteins, UPR/ERAD/EGAD, GCNA, ER, balbiani body, etc) makes the narrative of the paper very difficult to follow.

We are not jumping between topics, but following a narrative relevant to the central question of whether female mouse germ cells develop using a fusome. EMA, microtubule fragmentation, polarization proteins, ER, and balbiani body are all topics with a known connection to fusomes. This is explained in the general introduction and in relevant subsections. We appreciate this feedback that further explanations of these connections would be helpful. In the revised manuscript, use of the unified term mouse fusome will also help connect the narrative across sections. UPR/ERAD/EGAD are processes that have been studied in repair and maintenance of somatic cells and in yeast meiosis. We show that the major regulator XbpI is found in the fusome, and that the fusome and these rejuvenation pathway genes are expressed and maintained throughout oogenesis, rather than only during limited late stages as suggested in previous literature.

(7) The heading title "Visham participates in organelle rejuvenation during meiosis" in line 241 is speculative and/or not supported. Drawing upon the extensive, highly rigorous *Drosophila* literature, it is safe to extrapolate, but the claim about regeneration is not adequately supported.

We believe this statement is accurate given the broad scope of the term "participates." It is supported by localization of the UPR regulator XbpI to the fusome. XbpI is the ortholog of HacI a key gene mediating UPR-mediated rejuvenation during yeast meiosis. We also showed that rejuvenation pathway genes are expressed throughout most of meiosis (not previously known) and expanded cytological evidence of stage-specific organelle rejuvenation later in meiosis, such as mitochondrial-ER docking, in regions enriched in fusome antigens. However, we recognize the current limitations of this evidence in the mouse, and want to appropriately convey this, without going to what we believe would be an unjustified extreme of saying there is no evidence.

Reviewer #2 (Public Review):

We thank the reviewer for the comprehensive summary and for highlighting both the technical achievement and biological relevance of our study. We greatly appreciate the thoughtful suggestions that have helped us refine our presentation and terminology.

(1) Some titles contain strong terms that do not fully match the conclusions of the corresponding sections.

(1a) Article title "Mouse germline cysts contain a fusome-like structure that mediates oocyte development"

We will change the statement to: "Mouse germline cysts contain a fusome that supports germline cyst polarity and rejuvenation."

(1b) Result title "Visham overlaps centrosomes and moves on microtubules" We acknowledge that "moves" implies dynamics. We will include additional supplementary images showing

small vesicular components of the mouse fusome on spindle-derived microtubule tracks.

(1c) Result title “Visham associates with Golgi genes involved in UPR beginning at the onset of cyst formation”

We will revise this title to: “The mouse fusome associates with the UPR regulatory protein Xbp1 beginning at the onset of cyst formation” to reflect the specific UPR protein that was immunolocalized.

(1d) Result title “Visham participates in organelle rejuvenation during meiosis”

We will revise this to: “The mouse fusome persists during organelle rejuvenation in meiosis.”

(2) The authors aim to demonstrate that Visham is a fusome-like structure. I would suggest simply referring to it as a "fusome-like structure" rather than introducing a new term, which may confuse readers and does not necessarily help the authors' goal of showing the conservation of this structure in Drosophila and Xenopus germ cells. Interestingly, in a preprint from the same laboratory describing a similar structure in Xenopus germ cells, the authors refer to it as a "fusome-like structure (FLS)" (Davidian and Spradling, BioRxiv, 2025).

We appreciate the reviewer’s insightful comment. To maintain conceptual clarity and align with existing literature, we will refer to the structure as the mouse fusome throughout the manuscript, avoiding introduction of a new term.

Reviewer #3 (Public Review):

We thank the reviewer for emphasizing the importance of our study and for providing constructive feedback that will help us clarify and strengthen our conclusions.

(1) Line 86 - the heading for this section is "PGCs contain a Golgi-rich structure known as the EMA granule"

We agree that the enrichment of Golgi within the EMA PGCs was not shown until the next section. We will revise this heading to:

“PGCs contain an asymmetric EMA granule.”

(2) Line 105-106, how do we know if what's seen by EM corresponds to the EMA1 granule?

We will clarify that this identification is based on co-localization with Golgi markers (GM130 and GS28) and response to Brefeldin A treatment, which will be included as supplementary data. These findings support that the mouse fusome is Golgi-derived and can therefore be visualized by EM. The Golgi regions in E13.5 cyst cells move close together and associate with ring canals as visualized by EM (Figure 1E), the same as the mouse fusomes identified by EMA.

(3) Line 106-107-states "Visham co-stained with the Golgi protein Gm130 and the recycling endosomal protein Rab11a1". This is not convincing as there is only one example of each image, and both appear to be distorted.

Space is at a premium in these figures, but we have no limitation on data documenting this absolutely clear co-localization. We will replace the existing images with high-resolution, non-compressed versions for the final figures to clearly illustrate the co-staining patterns for GM130 and Rab11a1.

(4) Line 132-133---while visham formation is disrupted when microtubules are disrupted, I am not convinced that visham moves on microtubules as stated in the heading of this section.

We will include additional supplementary data showing small mouse fusome vesicles aligned along microtubules.

(5) Line 156 - the heading for this section states that Visham associates with polarity and microtubule genes, including pard3, but only evidence for pard3 is presented.

We agree and will revise the heading to: “Mouse fusome associates with the polarity protein Pard3.” We are adding data showing association of small fusome vesicles on microtubules.

(6) Lines 196-210 - it's strange to say that UPR genes depend on DAZ, as they are upregulated in the mutants. I think there are important observations here, but it's unclear what is being concluded.

UPR genes are not upregulated in DAZ in the sense we have never documented them increasing. We show that UPR genes during this time behave like pluripotency genes and normally decline, but in DAZ mutants their decline is slowed. We will rephrase the paragraph to clarify that Dazl mutation partially decouples developmental processes that are normally linked, which alters UPR gene expression relative to cyst development.

(7) Line 257-259-wave 1 and 2 follicles need to be explained in the introduction, and how these fits with the observations here clarified.

Follicle waves are too small a focus of the current study to explain in the introduction, but we will request readers to refer to the cited relevant literature (Yin and Spradling, 2025) for further details.

We sincerely thank all reviewers for their insightful and constructive feedback. We believe that the planned revisions—particularly the refined terminology, improved image quality, clarified statistics, and restructured abstract—will substantially strengthen the manuscript and enhance clarity for readers.

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